AI/ML DRIVEN MAINTENANCE SERVICE PROVIDER MATCHING PLATFORM FOR SRI LANKA

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Online service marketplaces improve access to service providers, but selecting a provider who closely matches a user's requirement remains difficult when platforms depend on keyword search, category filtering, ratings, or interaction history alone. This individual research component proposes a Hybrid Recommendation Framework for Intelligent Service Provider Matching. The framework combines TF-IDF content matching, multilingual sentence-embedding-based semantic similarity, and a custom credibility-based collaborative filtering approach. Provider credibility is estimated from average rating, booking success rate, and interaction frequency, while a user-history mechanism provides an additional personalization boost for previously interacted providers. The three component scores are normalized and fused using weights of 0.30 for TF-IDF, 0.35 for semantic matching, and 0.35 for collaborative filtering. The framework was evaluated using datasets containing 10,000 providers, 20,000 user requests, and 100,000 user-provider interactions. The hybrid approach achieved Precision@5, Precision@10, and Precision@20 values of 0.9490, 0.9483, and 0.9442, respectively, with NDCG@5, NDCG@10, and NDCG@20 values of 0.9496, 0.9490, and 0.9458. These results were higher than the individual TF-IDF, semantic, and collaborative-filtering configurations in the reported evaluation. The individual contribution of this work is the design and integration of a credibility-aware collaborative filtering mechanism within a modular hybrid recommendation pipeline for service-provider ranking.

Keywords: Hybrid recommender systems, service provider matching, TF-IDF, multilingual BERT, collaborative filtering

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TABLE OF CONTENTS

LIST OF FIGURES ix

LIST OF TABLES x

LIST OF ABBREVIATIONS xi

1 INTRODUCTION 1

1.1 Overall project context 1

1.2 Individual component context 2

1.3 Problem statement 3

1.4 Research gap 4

1.5 Aim and objectives 5

1.6 Research questions or hypotheses 6

1.7 Scope limitations and assumptions 7

1.8 Individual contribution 8

1.9 Thesis structure 9

2 LITERATURE REVIEW AND RELATED WORK 11

2.1 Review method and scope 11

2.2 Domain and theoretical foundations 12

2.3 Related research 13

2.4 Existing systems products and techniques 14

2.5 Gap synthesis and conceptual framework 16

3 RESEARCH METHODOLOGY 18

3.1 Research design 18

3.2 Data participants or study materials 20

3.3 Experimental or study procedure 21

3.4 Measures and evaluation metrics 23
3.5 Data analysis 24
3.6 Ethics privacy security and responsible AI 26
3.7 Reproducibility and research artefacts 28
4 PRODUCT AND SOLUTION DESIGN 31
4.1 Requirements 31
4.1.1 Functional Requirements 31
4.1.2 Non-Functional Requirements 32
4.2 Component architecture within the complete solution 33
4.3 Design alternatives and decisions 34
4.3.1 TF-IDF as a Standalone Approach 34
4.3.2 BERT as a Standalone Approach 35
4.3.3 Traditional Collaborative Filtering [5], [6] 35
4.3.4 4.3.4 Selection of the Hybrid Architecture 35
4.4 Detailed design 36
4.4.1 Recommendation Processing Workflow 36
4.4.2 TF-IDF Component Design 36
4.4.2 Multilingual BERT Component Design 37
4.4.4 Credibility-Based Collaborative Filtering Design 37
4.4.5 Score Normalization and Hybrid Fusion 38
4.4.6 Provider Ranking and Output 39
4.5 Standards and professional practice 39
5 IMPLEMENTATION INTEGRATION AND TESTING 41
5.1 Development environment and technology stack 41
5.2 Component implementation 42
5.2.1 Data Preparation and Preprocessing 42

5.2.2 TF-IDF Implementation 43
5.2.3 Multilingual BERT Implementation 44
5.3 Integration with the other components 47
5.4 Verification and software testing 48
5.4.1 Unit and Functional Testing 48
5.4.2 Recommendation Performance Testing 49
5.5 Deployment and operation 50
6 RESULTS AND VALIDATION 52
6.1 Research results 52
6.1.1 Recommendation Performance 52
6.1.2 Score Distribution Analysis 53
6.1.3 Component Correlation Analysis 54
6.1.4 Hybrid Weight Sensitivity Analysis 55
6.2 Product and technical results 56
6.3 User or stakeholder validation 57
6.4 Integration results 58
6.5 Summary of findings 59
7 DISCUSSION 61
7.1 Interpretation of findings 61
7.2 Comparison with literature baselines and existing solutions 62
7.3 Research contribution 64
7.4 Product contribution 65
7.5 Limitations and threats to validity 67
7.5.1 Category-Based Ground Truth 67
7.5.2 Limited Evaluation of Credibility 67
7.5.3 Cold-Start and Data Sparsity 67

7.5.4 Dataset and Generalizability Limitations 68
7.5.5 Heuristic Weight Selection 68
7.5.6 Potential Bias in Historical Information 69
7.5.7 Limited Real-World User Validation 69
7.6 Implications 70
8 CONCLUSION AND FUTURE DIRECTIONS 72
8.1 Conclusion 72
8.2 Achievement of objectives 73
8.3 Individual research and product contributions 75
8.4 Future directions 76
8.4.1 Adaptive Weight Optimization 76
8.4.2 Improved Cold-Start Handling 77
8.4.3 Real-World User and Provider Validation 77
8.4.4 Enhanced Credibility and Trust Modelling 77
8.4.5 Multilingual and Context-Aware Recommendation 78
8.4.6 Scalability and Deployment Optimization 78
8.4.7 Fairness and Explainable Recommendations 79
9 REFERENCES 80
10 GLOSSARY 82
11 APPENDICES 84
11.1 Appendix A Research instruments or extended results 84
11.1.1 A.1 Dataset Structure and Attributes 84
11.1.2 A.2 Extended Experimental Results 84
11.1.3 A.3 Research Artefacts 85
11.2 Appendix B Academic integrity and AI use disclosure 86
11.2.1 B.1 Declaration of Academic Integrity 86

11.2.2 B.2 AI Tools Used 86

11.2.3 B.6 Limitations of AI Assistance 86

LIST OF FIGURES

Figure 1.1:Provider credibility scoring chart 9
Figure 3.1:Research design flowchart 19
Figure 4.1:Architecture of the Hybrid Recommendation Component 34
Figure5.1:Development Environment Used for Recommendation Model Implementation 42
Figure 5.2:Sample of Preprocessed Provider Data 43
Figure 5.3:TF-IDF output 44
Figure 5.4:BERT based semantic matching 45
Figure 5.5:Sample output of CF 46
Figure 5.6:Sample Output of Hybrid Score Generation 47
Figure 5.7:Integrated Recommendation Output Generated by the System 50
Figure 6.1:Recommendation Performance Comparison 52
Figure 6.2:Distribution of TF-IDF, BERT and Collaborative Filtering Scores 54
Figure 6.3:Pearson Correlation between Recommendation Component Scores 55
Figure 6.4:Effect of Hybrid Weight Configurations 55
Figure 6.5:Ranked Provider Output 57
Figure 6.6:Integrated Service Provider Recommendation Output Presented through the Application 59

LIST OF TABLES

Table 2.1:Comparison of Existing Techniques for Service-Provider Recommendation 15
Table 3.1:Experimental procedure 22
Table 4.1:Functional Requirements of the Recommendation 31
Table 4.2:Non-functional requirements 32
Table 4.3: Comparison of Design Alternatives 35
Table 4.4:Credibility Factors 38
Table 4.5:Final Hybrid Score Configuration 38
Table 5.1:Development Environment and Technology Stack 41
Table 5.2:Functional test cases 48
Table 8.1: Achievement of Research Objectives 74
Table 11.1:Main Dataset Attributes 84
Table 11.2:Detailed Recommendation Performance 85
Table 11.3:AI Tool Usage Disclosure 86

LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
CBF	Content-Based Filtering
CF	Collaborative Filtering
IR	Information Retrieval
ML	Machine Learning
NDCG	Normalized Discounted Cumulative Gain
NLP	Natural Language Processing
TF-IDF	Term Frequency-Inverse Document Frequency
UI	User Interface

1 INTRODUCTION

1.1 Overall project context

Digital service marketplaces connect customers with skilled professionals who provide a wide range of essential household, commercial, and technical services, including plumbing, electrical work, carpentry, masonry, air-conditioning maintenance, cleaning, painting, CCTV installation, and vehicle repair. With the increasing use of digital platforms, customers are gradually moving away from purely traditional methods of finding service providers and are seeking faster and more convenient ways to identify suitable professionals. However, the central value of such a platform is not merely the availability of a large number of provider listings. A more important requirement is the platform's ability to intelligently guide users toward service providers who are relevant to their specific needs, credible in terms of their service history and reputation, and suitable for the stated service requirement.

In the Sri Lankan context, customers commonly discover service providers through classified advertisements, social-media posts and groups, personal recommendations, online directories, and general web searches. Although these channels provide access to a considerable number of service providers, the process of selecting the most appropriate provider often remains largely manual. Users may need to examine service categories, provider descriptions, skills, experience, ratings, reviews, location-related information, availability, and other relevant attributes individually before making a decision. This process can become time-consuming and difficult, particularly when several providers offer similar services or when users have limited information for determining the reliability and suitability of each provider.

Furthermore, a simple keyword-based search may not always accurately represent the actual service requirement of a customer. Different users may describe similar maintenance problems using different words, while a single service category may contain professionals with different levels of expertise, experience, credibility, and specialization. Therefore, identifying an appropriate provider requires consideration of several factors simultaneously rather than relying only on direct keyword

matching or overall ratings. The complete research project consequently considers service-provider matching as a multi-dimensional decision and recommendation problem rather than a conventional search problem.

The overall research project proposes an AI/ML-driven maintenance service-provider matching platform designed to improve the process of connecting customers with appropriate professionals. The complete system consists of multiple intelligent components that address different aspects of the provider-selection process, including personalized recommendation, contextual service matching, skill and credibility verification, and intelligent review analysis. Together, these components are intended to reduce the effort required from users when evaluating providers and to support more informed and reliable service-selection decisions.

1.2 Individual component context

The individual component developed in this work is a Hybrid Recommendation Framework for Intelligent Service Provider Matching. Its main responsibility is to process a natural-language service request and rank candidate service providers according to multiple recommendation signals. Rather than relying on a single matching technique, the framework combines different sources of information to produce more relevant and reliable recommendations.

The component integrates lexical relevance, semantic relevance, and interaction-based personalization. TF-IDF is used to preserve explicit keyword-level evidence between the user request and provider information, while a multilingual sentence-transformer model is used to capture contextual and semantic similarities that may not be identified through direct keyword matching. In addition, a custom credibility-based collaborative-filtering mechanism incorporates provider-quality information and previous user interactions into the recommendation process. [3], [4], [14]

Since these techniques generate scores on different numerical scales, the individual scores are normalized before being combined. The normalized scores are then fused to calculate a final recommendation score for each candidate provider, which is used to generate the final ranked list.

The framework is designed using a modular architecture, where each recommendation technique produces an independent score before the fusion stage. This makes it possible to analyze the contribution of each technique, compare individual methods against the hybrid approach, and identify areas for improvement. Furthermore, individual modules can be modified, replaced, or extended without redesigning the complete recommendation pipeline, supporting future improvements to the proposed framework.

1.3 Problem statement

Existing service-provider discovery approaches can return a large number of potential candidates without adequately understanding the actual meaning of a user's service request or the reliability of the providers being recommended. In many cases, users are required to manually compare provider profiles, service descriptions, ratings, experience, and other information before selecting a suitable professional. This becomes increasingly difficult when a platform contains a large number of providers offering similar services.

Traditional keyword-based matching mainly depends on direct word overlap between a user request and provider information. As a result, it may fail to identify relevant providers when the same service requirement is expressed using different words or phrases. Similarly, interaction-based recommendation methods can become less effective when users have limited historical interactions or when sufficient behavioural data are not available. Furthermore, provider quality and credibility are important factors in service selection and should directly contribute to the ranking process rather than being presented only as additional profile information. [1]

Therefore, a more comprehensive service-provider recommendation mechanism is required that combines explicit content relevance, contextual semantic similarity, user interaction history, and provider credibility within a unified ranking process. The key problems addressed in this research can be summarized as follows:

- Large provider sets make manual comparison and selection time-consuming for users.

- Lexical matching alone cannot reliably identify relevant providers when service requirements are expressed using different wording.
- Interaction-based recommendation may provide limited personalization for users with sparse interaction histories.
- Provider quality and credibility should directly influence recommendation rankings rather than being treated only as display attributes.

1.4 Research gap

Classical information-retrieval techniques provide efficient lexical matching by identifying direct similarities between a user query and provider-related textual information. Modern transformer-based models extend this capability by generating semantic representations that capture contextual meaning even when different words are used to express similar service requirements. In addition, collaborative-filtering techniques provide personalization by learning from user–provider interaction patterns and previous behavioural information. However, each of these approaches captures a different aspect of the service-provider matching problem and may have limitations when applied independently. [1]

Lexical approaches are effective at preserving explicit keyword evidence but may fail when user requests and provider descriptions use semantically similar but different terminology. Semantic models can address this limitation by identifying contextual similarity, but semantic relevance alone does not consider previous user interactions or provider reliability. Similarly, traditional collaborative filtering can support personalized recommendations, but its effectiveness may decrease when interaction histories are sparse or when provider quality is not directly considered during ranking. [5], [6]

The research gap addressed in this study is therefore the absence, within the target component, of a unified ranking mechanism that simultaneously preserves lexical relevance, captures multilingual semantic similarity, and incorporates a credibility-aware personalization signal. The proposed approach investigates how these complementary recommendation signals can be combined within a single hybrid framework to provide a more balanced ranking of service providers.

In particular, the research investigates a practical credibility-based extension to interaction-driven recommendation by incorporating provider rating, booking success, and interaction frequency. These factors allow provider credibility and previous user behaviour to contribute directly to the recommendation score rather than being treated only as additional profile information. By integrating this credibility-aware signal with lexical and semantic relevance, the research aims to address the limitations of individual recommendation techniques while improving the overall suitability of service-provider rankings.

1.5 Aim and objectives

The main aim of this research is to design, implement, and evaluate a hybrid recommendation framework that improves intelligent matching between natural-language service requests and suitable service providers. The framework aims to combine lexical relevance, semantic understanding, user interaction patterns, and provider credibility within a unified recommendation process to generate more relevant and reliable provider rankings.

To achieve the above aim, the following research objectives are defined:

1. Develop a TF-IDF-based content-matching module to measure explicit lexical relevance between natural-language service requests and service-provider information.
2. Integrate multilingual semantic embeddings using a sentence-transformer model to capture contextual similarities between user requests and provider descriptions, including cases where different wording is used to express similar service requirements.
3. Develop a credibility-based collaborative-filtering mechanism that incorporates provider-quality indicators and user interaction signals, including provider ratings, booking success, and interaction frequency, to support personalized recommendations.
4. Combine the individual recommendation scores through appropriate normalization and weighted score fusion to generate a unified final ranking score for each candidate service provider.

5. Evaluate the proposed hybrid recommendation framework against its individual recommendation components using suitable ranking and recommendation metrics to determine the contribution and effectiveness of the combined approach.
6. Integrate the developed recommendation component with the application backend, Firebase data layer, and user-facing workflow to demonstrate its practical implementation within the overall service-provider matching platform.

1.6 Research questions or hypotheses

Based on the identified research problem, research gap, and objectives, this study investigates how multiple recommendation techniques can be combined to improve intelligent service-provider matching. The research focuses on lexical relevance, semantic understanding, credibility-aware personalization, and the overall effectiveness of the proposed hybrid recommendation framework.

The following research questions guide the study:

RQ1: How effectively can TF-IDF-based content matching identify service providers based on the explicit lexical relevance of natural-language service requests?

RQ2: To what extent can multilingual semantic embeddings improve provider matching when user requests and provider descriptions express similar meanings using different words or languages?

RQ3: How can provider credibility factors, including provider ratings, booking success, and interaction frequency, be incorporated into collaborative filtering to improve personalized service-provider recommendations?

RQ4: Does combining lexical, semantic, and credibility-based collaborative-filtering scores through normalization and weighted score fusion improve provider ranking compared with the individual recommendation components?

RQ5: How does the proposed hybrid recommendation framework perform in terms of relevant ranking metrics when compared with TF-IDF, semantic matching, and collaborative filtering as individual baseline approaches?

RQ6: How effectively can the proposed recommendation framework be integrated with the application backend, Firebase data layer, and user-facing workflow for practical service-provider recommendation?

These research questions provide a structured basis for the design, implementation, and evaluation of the proposed hybrid recommendation framework. They also support the assessment of both the individual recommendation techniques and the overall contribution of their integration within a unified provider-ranking process.

1.7 Scope limitations and assumptions

The scope of this thesis is limited to the recommendation component of the overall service-provider matching platform. It covers data preparation and processing, TF-IDF-based lexical matching, multilingual semantic matching, credibility-based collaborative filtering, score normalization, hybrid score fusion, provider ranking, system integration, and evaluation of the proposed recommendation framework.

The recommendation component is integrated with the application backend and Firebase database, allowing users and service providers to register and interact with the platform**.** Service providers can create and maintain their profiles and service-related information, while registered users can submit natural-language service requests and interact with providers. Therefore, the recommendation process is not limited to a fixed dataset; it can operate using provider and interaction information stored and updated within the application database. [12]

For experimental evaluation, prepared and controlled data are used where necessary to systematically compare the individual recommendation techniques and the proposed hybrid framework. These data support repeatable evaluation and measurement of ranking performance. However, the implemented system is designed to process dynamically stored provider information and user interactions as the platform is used.

The thesis specifically focuses on recommendation and provider ranking. Features such as contextual availability, weather-based information, detailed skill verification, and review-sentiment analysis are handled by other components of the overall

research project and are therefore outside the primary experimental scope of this study.

A limitation of the current research is that the recommendation framework has not been evaluated through a large-scale, long-term production deployment involving a continuously growing population of real users and service providers. Therefore, the evaluation demonstrates the performance and behaviour of the proposed framework within the implemented system and experimental conditions, while further real-world deployment could be used to evaluate its performance at a larger scale.

1.8 Individual contribution

The principal individual contribution of this research is the development of a custom credibility-based collaborative-filtering mechanism for intelligent service-provider recommendation and its integration into a hybrid ranking framework. Unlike a conventional interaction-based approach that mainly relies on previous user behaviour, the proposed mechanism introduces provider credibility directly into the recommendation process. Provider credibility is calculated using three main signals: normalized average rating, booking success rate, and transformed interaction frequency. These factors are combined to generate a credibility score for each provider, allowing providers with stronger service quality and interaction records to receive an appropriate influence during recommendation.

The mechanism further incorporates user interaction history to provide controlled personalization**.** When a user has previously interacted with a provider, the provider's credibility-based score receives a defined loyalty boost, while new users without an interaction history continue to receive recommendations based on the general credibility scores. The resulting values are cleaned, normalized, and maintained within the valid score range before being passed to the hybrid recommendation pipeline. This approach helps address sparse user-history situations while still introducing personalization when relevant interaction data are available.

A further contribution is the integration of this credibility-aware collaborative score with TF-IDF lexical relevance and multilingual semantic similarity. Rather than allowing a single recommendation method to determine the ranking, each technique

independently evaluates a different aspect of provider suitability. The individual scores are subsequently normalized and combined through weighted score fusion to produce the final provider ranking. This modular design also enables the contribution of the credibility-based mechanism to be evaluated independently and compared with the other recommendation components.

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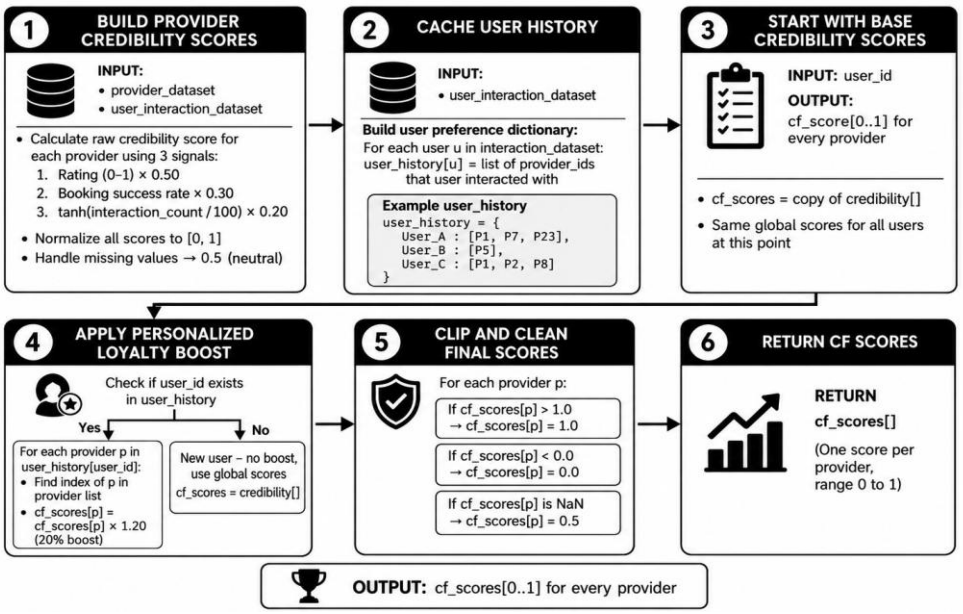


Figure 1.1:Provider credibility scoring chart

1.9 Thesis structure

This thesis is organized into eight chapters, with each chapter addressing a specific stage of the research and development process. Chapter 2 reviews the relevant literature, existing recommendation approaches, and technical foundations related to content-based filtering, semantic matching, collaborative filtering, and hybrid recommendation systems. It also establishes the theoretical background required for the proposed research.

Chapter 3 describes the research methodology, including the overall research approach, data preparation, recommendation techniques, experimental design, and evaluation procedure. Chapter 4 presents the product and solution design, including

the proposed system architecture, recommendation workflow, component design, and integration with the overall service-provider platform.

Chapter 5 explains the implementation of the proposed recommendation framework, its integration with the backend and database, and the testing procedures used to verify the developed functionality. Chapter 6 presents the experimental and technical results, including the performance of individual recommendation methods and the final hybrid framework using the selected evaluation metrics.

Chapter 7 discusses the findings, contributions, and limitations of the framework. Finally, Chapter 8 concludes the thesis by summarizing the research outcomes and identifying future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review focuses on the main research areas that support the development of the proposed hybrid service-provider recommendation framework. These areas include information retrieval, content-based recommendation, semantic text representation, collaborative filtering, hybrid recommender systems, provider credibility, and ranking evaluation. The primary purpose of the review is to identify suitable techniques for matching natural-language service requests with relevant service providers and to establish the theoretical and technical foundation for the proposed hybrid architecture.

The review considers both foundational techniques and recent developments in recommendation and natural-language processing. Classical information-retrieval approaches are examined to understand how lexical similarity and keyword relevance can be measured using techniques such as TF-IDF. In addition, transformer-based sentence representation methods are reviewed to investigate how semantic similarity can capture the contextual meaning of service requests and provider descriptions, particularly when similar requirements are expressed using different wording or languages.

Collaborative-filtering literature is also examined to understand how historical user interactions can support personalized recommendations. Particular attention is given to the limitations of interaction-based approaches when user histories are sparse and to the potential role of provider credibility and quality indicators in strengthening recommendation scores. Hybrid recommender systems are then reviewed to determine how lexical, semantic, and interaction-based signals can be combined within a unified ranking process.

Finally, the review considers commonly used ranking and recommendation evaluation methods to establish an appropriate basis for comparing the individual recommendation components with the proposed hybrid framework. Overall, the literature review is limited to concepts and techniques directly relevant to the individual recommendation component of this thesis, while contextual availability,

weather analysis, detailed skill verification, and review-sentiment analysis are outside its primary scope.

2.2 Domain and theoretical foundations

Digital service-provider marketplaces operate as two-sided platforms that connect customers who require specific services with professionals capable of providing them. Unlike conventional product recommendation, service-provider matching involves selecting a suitable individual or business based on multiple factors such as service relevance, provider expertise, previous interactions, ratings, credibility, and user requirements. Therefore, effective provider recommendation requires more than simply retrieving profiles that contain matching keywords.

From an information-retrieval perspective, a user's natural-language service request can be treated as a query, while provider profiles and service descriptions represent candidate documents. The recommendation task therefore involves estimating the relevance of each provider to the submitted request and ranking the candidates accordingly. Lexical matching provides one theoretical foundation for this process by measuring direct term overlap between the request and provider information. TF-IDF is commonly used for this purpose because it assigns greater importance to informative terms while reducing the influence of frequently occurring terms. [1]

However, lexical similarity alone may not capture the complete meaning of a service request. Users may describe the same problem using different words, expressions, or languages. Semantic representation addresses this limitation by representing sentences as numerical embeddings in a shared vector space. Semantically related requests and provider descriptions can therefore remain similar even when they do not contain identical keywords. This principle provides the theoretical basis for incorporating multilingual sentence embeddings into the proposed framework. [4]

Another important foundation is collaborative recommendation, where previous user interactions are used to introduce personalization. In a service marketplace, interaction history can provide evidence about previous relationships or preferences between users and providers. However, interaction information alone may be insufficient, particularly for new users or users with limited histories. For this reason,

the proposed research extends the interaction-based concept by incorporating provider credibility, derived from indicators such as ratings, booking success, and interaction frequency. [5], [6]

The overall framework is based on the principle of hybrid recommendation, where multiple complementary signals are combined to overcome the limitations of individual techniques. In this research, lexical relevance, semantic similarity, and credibility-aware personalization represent different dimensions of provider suitability. Their normalized scores are combined through weighted fusion to generate a final ranking, providing the theoretical basis for the proposed intelligent service-provider matching framework. [7], [8]

2.3 Related research

Previous research on recommender systems has explored several approaches for improving the relevance and personalization of recommendations. Content-based recommendation methods generally compare user requirements with item or provider characteristics and recommend alternatives with similar features. In text-oriented applications, techniques such as TF-IDF and cosine similarity have been widely applied to measure lexical similarity. These approaches are computationally efficient and provide interpretable evidence based on shared terms. However, their dependence on word overlap can reduce their effectiveness when the same requirement is expressed using different terminology. [1], [6]

Recent research has increasingly applied transformer-based language models and sentence embeddings to overcome the limitations of traditional lexical matching. Sentence-transformer architectures represent complete sentences as dense numerical vectors, enabling similarity to be measured according to contextual meaning rather than exact word overlap. Multilingual embedding models further extend this capability by representing text from multiple languages within a comparable semantic space. Such techniques are particularly relevant to service-provider marketplaces where users may describe similar maintenance requirements using different expressions or languages. [3], [4]

Another major research direction is collaborative filtering, which generates personalized recommendations from historical user interactions. Collaborative approaches can identify preference patterns that are not directly available from textual content. Nevertheless, their performance can be affected by data sparsity and cold-start conditions, particularly when new users or providers have limited interaction histories. Research on recommender systems has therefore investigated the use of additional signals, such as ratings, popularity, trust, reputation, and other quality indicators, to improve recommendation reliability. [5], [6], [10]

Hybrid recommender systems combine two or more recommendation techniques to reduce the limitations of individual methods. Existing studies have demonstrated different hybrid strategies, including weighted score combination, switching mechanisms, feature-level integration, and model-level fusion. Weighted hybrid approaches are especially suitable when independent recommendation components generate comparable scores, since each signal can be normalized and assigned an appropriate contribution to the final ranking. [7], [8]

Building on these research directions, the present study combines TF-IDF-based lexical matching, multilingual semantic similarity, and credibility-aware interaction-based recommendation within a single hybrid framework. The proposed work differs from using these techniques independently by introducing a custom credibility-based collaborative mechanism that considers provider rating, booking success, interaction frequency, and user history. The resulting score is then combined with lexical and semantic relevance to generate the final service-provider ranking. This allows the research to investigate whether complementary recommendation signals can provide more effective provider matching than individual recommendation approaches.

2.4 Existing systems products and techniques

Existing digital service marketplaces provide different mechanisms for connecting customers with professionals such as plumbers, electricians, cleaners, technicians, and other maintenance-service providers. Most platforms allow users to search or browse providers according to service categories, location, ratings, availability, and profile information. These mechanisms make provider discovery easier than

traditional methods such as personal referrals or classified advertisements. However, the level of intelligence used to understand a user's specific natural-language requirement can vary considerably between platforms.

Many existing systems primarily depend on category-based filtering and keyword search. A customer selects a predefined service category or enters a search term, after which the system retrieves providers associated with that category or keyword. This approach is simple and efficient but places considerable responsibility on the user to select the correct category and manually compare the returned providers. Keyword-oriented retrieval can also become less effective when the terminology used in a customer request differs from the wording contained in provider profiles.

Provider reputation is another important feature of existing service marketplaces. Ratings, reviews, completed jobs, and similar indicators are commonly presented to users to support provider selection. However, displaying credibility information separately requires users to consider it manually after providers have already been retrieved. Incorporating such information directly into the recommendation score provides an opportunity for provider quality to influence the ranking itself.

Table 2.1:Comparison of Existing Techniques for Service-Provider Recommendation

Technique	**Main Purpose**	**Key Strength**	**Main Limitation**
TF-IDF Matching	Lexical matching	Captures explicit keyword relevance	Limited semantic understanding
Semantic Matching	Context-aware matching	Understands similar meanings and different wording	Does not consider user history
Collaborative Filtering	Personalized recommendation	Uses previous user interactions	Limited by sparse interaction data
Credibility-Based	Provider-quality	Considers	May not capture

Ranking	assessment	provider reliability	request relevance
Hybrid Recommendation	Combined provider ranking	Integrates multiple recommendation signals	Requires effective score normalization and fusion

The comparison demonstrates that each recommendation technique addresses a different aspect of service-provider matching. Lexical and semantic approaches mainly focus on request relevance, while collaborative filtering and credibility-based ranking introduce personalization and provider-quality information. Therefore, the proposed research adopts a hybrid recommendation approach that combines these complementary signals to generate a more balanced and relevant ranking of service providers.

2.5 Gap synthesis and conceptual framework

The literature review indicates that existing recommendation techniques provide useful capabilities for service-provider matching, but each approach also introduces specific limitations. Lexical matching is effective for identifying explicit keyword similarities but has limited ability to understand requests expressed using different terminology. Semantic matching improves contextual understanding but does not independently consider user interaction history or provider credibility. Similarly, collaborative filtering supports personalization through previous interactions, but its effectiveness can be reduced when user histories are sparse. [1], [4], [6], [7]

Another important gap is the treatment of provider credibility within the recommendation process. Provider ratings, successful bookings, and interaction information can provide useful evidence about provider reliability, but such information is often considered separately from request relevance. For service-provider marketplaces, both relevance and credibility are important because a provider should not only match the requested service but should also demonstrate sufficient evidence of reliability.

Based on these identified gaps, this research proposes a conceptual framework that combines three complementary recommendation signals: TF-IDF-based lexical relevance, multilingual semantic similarity, and credibility-based collaborative filtering. Each component generates an independent score representing a different dimension of provider suitability. These scores are normalized and combined through weighted score fusion to generate a final recommendation score and ranked provider list.

This framework provides the foundation for the methodology and implementation presented in the following chapters. It also enables the individual recommendation components to be evaluated separately and compared with the complete hybrid approach, allowing the research to determine whether combining complementary recommendation signals improves service-provider ranking.

3 RESEARCH METHODOLOGY

3.1 Research design

This research adopts a design-and-evaluation-oriented quantitative research approach to develop and assess the proposed Hybrid Recommendation Framework for Intelligent Service Provider Matching. The research involves designing the recommendation architecture, implementing its individual components, integrating them into the service-provider platform, and experimentally evaluating their effectiveness in ranking relevant providers.

The research design is structured around three main recommendation components: TF-IDF-based lexical matching, multilingual semantic matching, and credibility-based collaborative filtering**.** Each component generates an independent recommendation score based on a different aspect of provider suitability. TF-IDF measures explicit textual relevance, semantic embeddings capture contextual similarity between service requests and provider descriptions, and the credibility-based collaborative mechanism incorporates provider-quality indicators and previous user interactions.

A comparative experimental design is used to evaluate the proposed approach. The individual recommendation methods are treated as baseline components and their ranking performance is compared with the complete hybrid framework. Before fusion, the generated scores are normalized to ensure that outputs from different recommendation methods can be combined consistently. Weighted score fusion is then applied to calculate the final recommendation score and produce the ranked provider list.

The research also includes system-level implementation and integration. The recommendation framework communicates with the application backend and Firebase data layer, allowing registered users to submit natural-language service requests and enabling registered providers and their updated information to participate in the recommendation process. Controlled and prepared data are used where necessary during experimental evaluation to ensure repeatability, while the

implemented system itself is capable of operating with dynamically stored user, provider, and interaction information.

The overall research process follows an iterative sequence of problem identification, literature analysis, component design, implementation, integration, experimentation, evaluation, and refinement**.** This design enables both the technical functionality of the developed system and the recommendation performance of the proposed hybrid approach to be systematically examined.

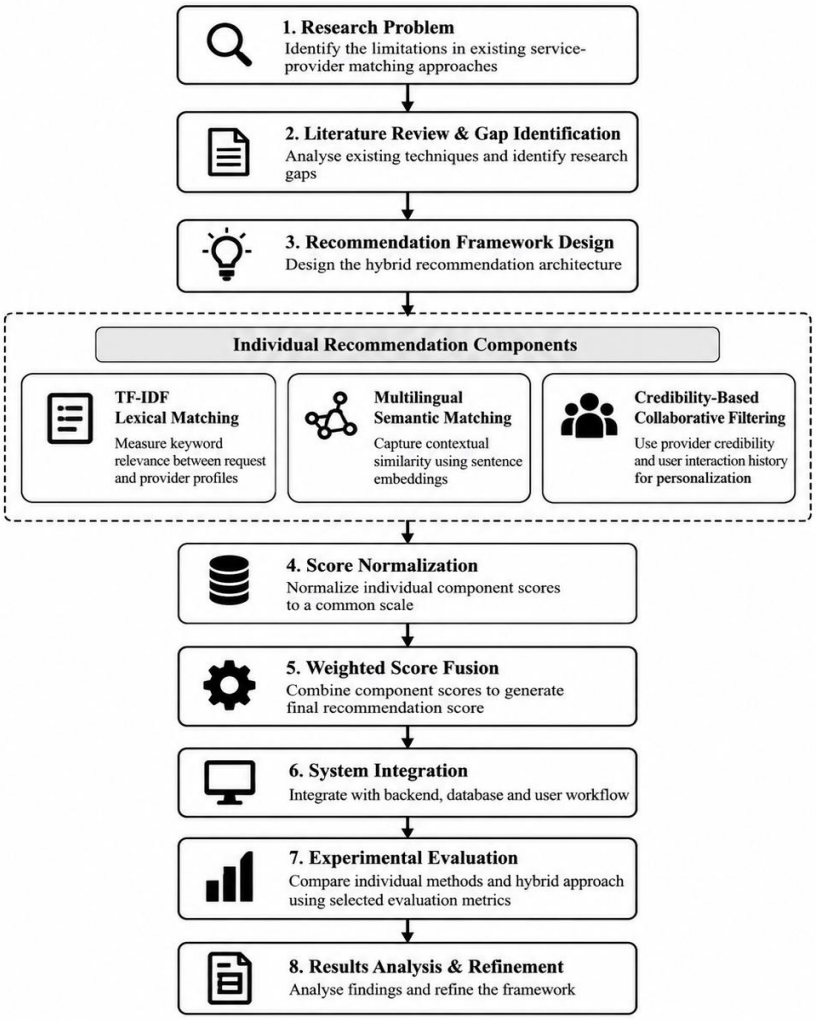


Figure 3.1:Research design flowchart

3.2 Data participants or study materials

The development and evaluation of the proposed hybrid recommendation framework require several types of data representing service providers, users, service requests, and user–provider interactions**.** The implemented application stores these data through the backend and firebase database, allowing registered users and service providers to create and update information as the system operates. Therefore, the recommendation framework is designed to work with dynamically stored application data rather than being restricted to a fixed dataset.

The provider data contain information required for matching and credibility calculation, including provider identifiers, service categories, service descriptions, ratings, booking-related information, and interaction counts. Textual provider information is used by the TF-IDF and semantic-matching components, while provider ratings, booking success, and interaction frequency contribute to the credibility-based collaborative-filtering mechanism.

The user and interaction data represent previous relationships between users and service providers. User identifiers and provider interaction histories are used to introduce personalization into the recommendation process. When sufficient interaction history exists, previously interacted providers can receive a controlled personalization boost. For new users or users with limited histories, the framework can continue generating recommendations using the available content, semantic, and provider-credibility signals.

Natural-language service requests are used as the primary input to the recommendation framework. These requests represent different ways in which customers may describe maintenance requirements, including variations in wording and language. The requests are processed by both lexical and multilingual semantic-matching modules before their scores are combined with the credibility-aware recommendation signal.

For controlled experimental evaluation, prepared test cases and structured records are used where necessary to ensure that the individual recommendation techniques and the complete hybrid framework can be compared consistently. These materials

support repeatable evaluation but do not restrict the operational system to static data. The implemented platform can process newly registered providers, updated provider information, new service requests, and user interactions stored through the application.

3.3 Experimental or study procedure

The experimental procedure was designed to evaluate both the individual recommendation techniques and the complete hybrid recommendation framework under consistent conditions. The procedure begins with the preparation of provider information, user interaction records, and representative natural-language service requests required by the recommendation components. Relevant textual and numerical attributes are then processed into suitable formats for lexical, semantic, and credibility-based recommendation.

For each service request, the TF-IDF module calculates lexical similarity between the request and available provider information. In parallel, the multilingual semantic-matching module generates sentence embeddings for the request and provider descriptions and calculates their semantic similarity. The credibility-based collaborative-filtering module independently generates provider scores using normalized provider ratings, booking success, interaction frequency, and available user interaction history.

Since the three recommendation components produce scores using different methods, the resulting values are normalized to a common range before fusion. The normalized scores are then combined using the defined weighted score-fusion mechanism to calculate a final hybrid score for each candidate provider. Providers are ordered according to these final scores to produce the recommendation ranking presented to the user.

The experimental evaluation is conducted by applying the same service requests and candidate-provider information to the individual recommendation methods and the complete hybrid framework. The resulting rankings are compared using the selected ranking metrics. This allows the experiment to determine whether combining lexical,

semantic, and credibility-aware personalization signals provides an improvement over using the individual recommendation components independently.

In addition to experimental performance, the complete recommendation workflow is tested after integration with the application backend and firebase database**.** This verifies that user requests can be processed, provider information can be retrieved from the application data layer, recommendation scores can be generated correctly, and ranked providers can be returned through the user-facing workflow.

Table 3.1:Experimental procedure

Stage	**Procedure**	**Output**
1. Data Preparation	Prepare provider, interaction and request data	Processed input data
2. Lexical Matching	Apply TF-IDF similarity	Lexical scores
3. Semantic Matching	Apply multilingual embeddings	Semantic scores
4. Credibility Scoring	Process credibility and interaction signals	CF scores
5. Score Fusion	Normalize and combine scores	Hybrid scores
6. Provider Ranking	Sort providers by final score	Ranked provider list
7. Evaluation	Compare individual and hybrid methods	Performance results
8. Integration Testing	Test complete application workflow	Verified system operation

3.4 Measures and evaluation metrics

The performance of the proposed Hybrid Recommendation Framework was evaluated using standard ranking-based Information Retrieval metrics. Since the objective of the proposed system is to generate a ranked list of the most relevant service providers for a given user request, Precision@K and Normalized Discounted Cumulative Gain (NDCG@K) were selected as the primary evaluation metrics. The evaluation was conducted at three recommendation cut-off levels, namely K=5, K=10 and K=20. [9], [10]

Precision@K measures the proportion of relevant service providers among the top K providers returned by the recommendation system. It is used to determine how accurately the system places relevant providers within the recommendation list. In this research, the service category associated with each user request was used as the ground-truth relevance label. A recommended provider was considered relevant when the provider's service category matched the service category associated with the corresponding user request. Therefore, a higher Precision@K value indicates that a larger proportion of the Top-K recommendations belong to the required service category. Precision was evaluated at K=5, K=10, and K=20 allowing the recommendation quality to be examined across both short and relatively larger recommendation lists. [10]

Although Precision@K measures the proportion of relevant providers in the recommendation list, it does not fully consider the positions at which those providers appear. In a recommendation system, providers appearing near the top of the ranked list are more important because users are more likely to examine higher-ranked recommendations. Therefore, NDCG@K was used to evaluate the quality of the ranking. Normalized Discounted Cumulative Gain (NDCG@K) was used to evaluate the ranking quality of the recommended service providers. Unlike Precision@K, which mainly considers whether relevant providers are included within the Top-K recommendations, NDCG@K also considers the position of each relevant provider in the ranked list. [9], [10]

In this evaluation, providers that matched the service category of the corresponding user request were considered relevant. When a relevant provider appeared near the

top of the recommendation list, it contributed more positively to the NDCG score. Relevant providers appearing at lower positions contributed less, while non-relevant providers did not improve the ranking quality. Therefore, this metric was useful for determining whether the recommendation methods placed the most relevant providers at higher positions rather than simply including them somewhere within the recommendation list.

The ranking obtained by each recommendation method was compared with an ideal ranking in which relevant providers were placed in the best possible positions. This comparison produced a normalized score between 0 and 1. A value closer to 1 represents a higher-quality ranking, indicating that relevant service providers are concentrated near the top of the recommendation list. In this study, NDCG was evaluated at K=5, K=10, and K=20 to assess ranking performance at different recommendation list sizes. [9]

For the experimental evaluation, 500 user requests were randomly sampled from the User Request Dataset using a fixed random seed to support reproducibility. For each request, providers were independently ranked using the TF-IDF, Multilingual BERT, Credibility-Based Collaborative Filtering, and Hybrid recommendation approaches.

The category associated with each user request was used to establish relevance. Providers belonging to the same service category as the request were considered relevant for the evaluation. Precision@K and NDCG@K were then calculated for each recommendation approach at K=5, (K=10, and K=20. The metric values were aggregated across the evaluated requests to compare the performance of the individual recommendation methods with the proposed Hybrid Recommendation Framework.

3.5 Data analysis

The data analysis process was conducted to examine the characteristics of the datasets and to evaluate the behaviour of the individual recommendation methods and the proposed Hybrid Recommendation Framework. The analysis focused on the Provider Dataset, User Request Dataset, and User Interaction Dataset, which provide

the information required for content-based, semantic, and credibility-based recommendation.

The Provider Dataset was analysed to identify information related to service categories, provider descriptions, locations, ratings, experience, and other provider-related attributes. Textual information such as service descriptions was particularly important for the TF-IDF and BERT components, since these methods compare user requests with provider information to determine relevance.

The User Request Dataset contains service requests representing different user requirements. These requests were used as the input queries for the recommendation process. During evaluation, 500 user requests were randomly selected using a fixed random seed to provide a consistent and reproducible evaluation sample. The service category associated with each request was used to determine the relevance of the recommended providers.

The User Interaction Dataset was analysed to obtain information about previous interactions between users and service providers. Provider ratings, successful booking outcomes, and interaction frequency were used by the credibility-based collaborative filtering component. These attributes were analysed together to represent different aspects of provider credibility, including service quality, reliability, and previous activity.

Before applying the recommendation algorithms, the datasets were preprocessed to handle missing or inconsistent values and to prepare textual and numerical attributes for further processing. The TF-IDF component analysed keyword-level similarity between user requests and provider descriptions, while the multilingual BERT component analysed semantic similarity. The credibility-based collaborative filtering component analysed historical interaction and provider credibility information. The scores generated by the three components were normalized to a common range before being combined by the hybrid recommendation method. [1], [13]

The recommendation results were analysed using Precision@K and NDCG@K at recommendation list sizes of 5, 10, and 20. Precision@K was used to examine how many category-relevant providers appeared within the Top-K recommendations,

while NDCG@K was used to examine whether relevant providers were positioned closer to the top of the ranked recommendation list.

A comparative analysis was also performed among the TF-IDF, BERT, Collaborative Filtering, and Hybrid approaches. This comparison was used to determine whether combining lexical relevance, semantic similarity, and credibility-based information improved recommendation performance compared with using each recommendation method independently.

In addition to the performance metrics, the generated recommendation scores were analysed to understand the contribution and behaviour of the individual components. Score distributions were examined to identify differences between TF-IDF, BERT, and collaborative filtering scores. Correlation analysis was used to investigate the relationship between the component scores, while weight sensitivity analysis was performed to examine the effect of different score combinations on the final hybrid recommendation performance. These analyses supported the selection and evaluation of the final hybrid configuration.

3.6 Ethics privacy security and responsible AI

The development of the Hybrid Recommendation Framework considered ethical, privacy, security, and responsible AI principles because the system processes user requests, provider information, ratings, booking outcomes, and historical interaction data. These considerations are important to ensure that the recommendation process is reliable, fair, and respectful of both users and service providers.

- **Ethical Considerations**

The recommendation system was designed to rank service providers based on relevant information rather than personal or sensitive characteristics. The recommendation process mainly considers service relevance, semantic similarity, historical interactions, and provider credibility indicators such as ratings, successful booking outcomes, and interaction frequency.

Care should also be taken to avoid unfairly disadvantaging new service providers who have limited historical interactions. Since historical information is used by the

credibility-based collaborative filtering component, providers with fewer previous interactions may receive less information-based support than established providers. The hybrid approach reduces dependence on interaction history by also considering TF-IDF and BERT-based content relevance.

- **Privacy**

The system requires user requests and historical interaction information to generate recommendations. Therefore, only information necessary for the recommendation process should be collected and processed. Personally identifiable or sensitive information should not be unnecessarily included in the recommendation calculations.

User interaction data should be handled only for its intended purpose, such as improving recommendation relevance and personalization. When datasets are used for research and evaluation, direct personal identifiers should be removed or anonymized where appropriate to reduce privacy risks.

- **Data Security**

Appropriate security measures should be applied when storing and accessing user, provider, and interaction data. Since Firebase is used as the database layer of the proposed system, database access should be restricted to authorized components and users. Authentication, controlled access permissions, secure communication, and appropriate credential management should be considered when deploying the system in a real-world environment.

Sensitive configuration information, such as database credentials and authentication details, should not be exposed through source code repositories or client-side applications. Regular backups and appropriate access controls can also help protect the integrity and availability of the system data.

- **Responsible AI**

Responsible AI principles were considered when designing the recommendation process. The proposed framework does not rely on a single recommendation signal. Instead, TF-IDF, multilingual BERT, and credibility-based collaborative filtering

provide different forms of evidence when generating the final ranking. This reduces complete dependence on a single recommendation technique.

The credibility score is based on observable service-related indicators, including provider ratings, booking success, and interaction frequency. However, these indicators may still contain biases originating from historical user behaviour or unequal levels of provider activity. Therefore, recommendation scores should be interpreted as decision-support information rather than an absolute judgement of a provider's quality.

Transparency is also important in responsible recommendation systems. The major factors influencing the recommendation process are identifiable within the proposed framework, including keyword relevance, semantic relevance, provider credibility, and historical interactions. This makes it possible to explain the general basis on which providers are ranked.

- **Fairness and Limitations**

Although the proposed system aims to provide relevant and credible service provider recommendations, it cannot guarantee that the highest-ranked provider will always be the best provider for every user or situation. Recommendation quality depends on the quality and completeness of the available data.

New providers with limited ratings or interaction histories may also be affected by the cold-start problem. Furthermore, historical ratings and interactions may reflect existing user preferences or biases. Future improvements could therefore include additional fairness-aware mechanisms, stronger cold-start handling, and continuous monitoring of recommendation outcomes to identify potential ranking bias.

3.7 Reproducibility and research artefacts

Reproducibility was considered throughout the development and evaluation of the proposed Hybrid Recommendation Framework. The implementation was organized so that the main recommendation process, dataset preparation, model execution, and evaluation procedures could be repeated under the same experimental conditions.

The experiments were implemented using Python with commonly available machine learning and data processing libraries, including Pandas, NumPy, Scikit-learn, and Sentence Transformers. Google Colab was used as the primary experimental environment, while Firebase was used for storing and managing provider and interaction-related data. The semantic matching component used the pretrained multilingual sentence-transformer model paraphrase-multilingual-MiniLM-L12-v2, allowing the same model configuration to be reused when reproducing the experiments.

To improve the consistency of the evaluation process, a fixed random seed was used when selecting the evaluation sample. A total of 500 user requests were randomly selected from the User Request Dataset for performance evaluation. Using a fixed random state ensures that the same evaluation sample can be selected when the experiment is executed again under the same dataset conditions.

The configuration of the hybrid recommendation method was also explicitly maintained. The final hybrid model combined TF-IDF, multilingual BERT, and credibility-based collaborative filtering using weights of 0.30, 0.35, and 0.35, respectively. The credibility calculation used provider rating, booking success rate, and interaction frequency with weights of 0.50, 0.30, and 0.20. Maintaining these configurations allows the recommendation scores and evaluation process to be repeated consistently.

The main research artefacts produced during this research include the implementation source code, Google Colab notebook, processed datasets, recommendation and evaluation scripts, experimental results, generated performance tables, and visualizations**.** The visual research artefacts include score distribution analysis, correlation analysis, and hybrid weight sensitivity analysis. These artefacts provide supporting evidence for the implementation and experimental findings presented in this thesis.

The recommendation methods were evaluated using the same evaluation procedure at K values of 5, 10, and 20, using Precision@K and NDCG@K. Applying a consistent evaluation procedure to TF-IDF, BERT, collaborative filtering, and the

proposed hybrid method supports a fair comparison between the different recommendation approaches.

Overall, maintaining the implementation code, model configurations, evaluation settings, and experimental outputs improves the reproducibility and traceability of the research. These artefacts can also support future extensions of the proposed framework and allow subsequent researchers to examine or repeat the experimental procedure under comparable conditions.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The requirements of the recommendation component were identified based on the main research problem. A user may describe a required service using natural language, while available service providers contain different descriptions, service categories, ratings, and historical interaction information. Therefore, the recommendation component must process these different sources of information and generate a ranked list of relevant providers.

4.1.1 Functional Requirements

Table 4.1:Functional Requirements of the Recommendation

Requirement	Description
Accept user request	The system should accept a natural-language service request from the user.
Retrieve provider data	The system should retrieve relevant provider information required for recommendation generation.
Keyword matching	The system should identify lexical similarity between user requests and provider descriptions using TF-IDF.
Semantic matching	The system should identify semantic similarity using multilingual BERT embeddings.
Credibility analysis	The system should evaluate provider credibility using ratings, booking success and interaction frequency.
Historical personalization	Previous user-provider interactions should be considered when available.
Score normalization	Scores generated by different recommendation

	components should be transformed into a comparable range.
Hybrid score generation	TF-IDF, BERT and collaborative filtering scores should be combined to generate a final ranking score.
Provider ranking	Providers should be ranked according to their final hybrid scores.
Top recommendations	The system should return the Top-20 providers for further processing or presentation.

4.1.2 Non-Functional Requirements

In addition to the functional requirements, several non-functional requirements were considered.

Table 4.2:Non-functional requirements

Requirement	**Description**
Performance	The recommendation component should generate ranked results within an acceptable response time.
Scalability	The architecture should support an increasing number of providers, requests, and interactions.
Reliability	The same inputs and configurations should produce consistent recommendation behaviour.
Maintainability	TF-IDF, BERT, and collaborative filtering modules should remain modular to support future modifications.
Security	Database credentials and sensitive configuration information should be protected from unauthorized access.

Privacy	Only information required for recommendation and personalization should be processed.
Fairness	The design should reduce complete dependence on historical interactions, particularly for providers with limited history.
Explainability	The main factors contributing to the recommendation score should remain identifiable.

4.2 Component architecture within the complete solution

The Hybrid Recommendation Framework operates as one component of the complete intelligent service provider matching platform. The overall platform contains multiple intelligent components responsible for recommendation generation, contextual matching, provider credibility and skill-related analysis, and review-related analysis.

When a user submits a service request through the application, the request is transferred through the backend to the recommendation component. The recommendation engine retrieves the required provider and interaction information from the database and processes the request through three recommendation paths.

The TF-IDF module evaluates direct lexical similarity between the user request and provider information. The multilingual BERT module evaluates contextual and semantic similarity, allowing the system to identify providers even when the wording of the request differs from provider descriptions. The credibility-based collaborative filtering module uses historical interactions and provider-level credibility information to provide an additional recommendation signal. [1], [13]

The outputs of these modules are normalized and combined by the hybrid score fusion layer. Providers are then ranked according to their final hybrid scores, and the Top-20 candidates are generated. These recommendation results can subsequently interact with other components of the complete platform before the final results are presented to the user.

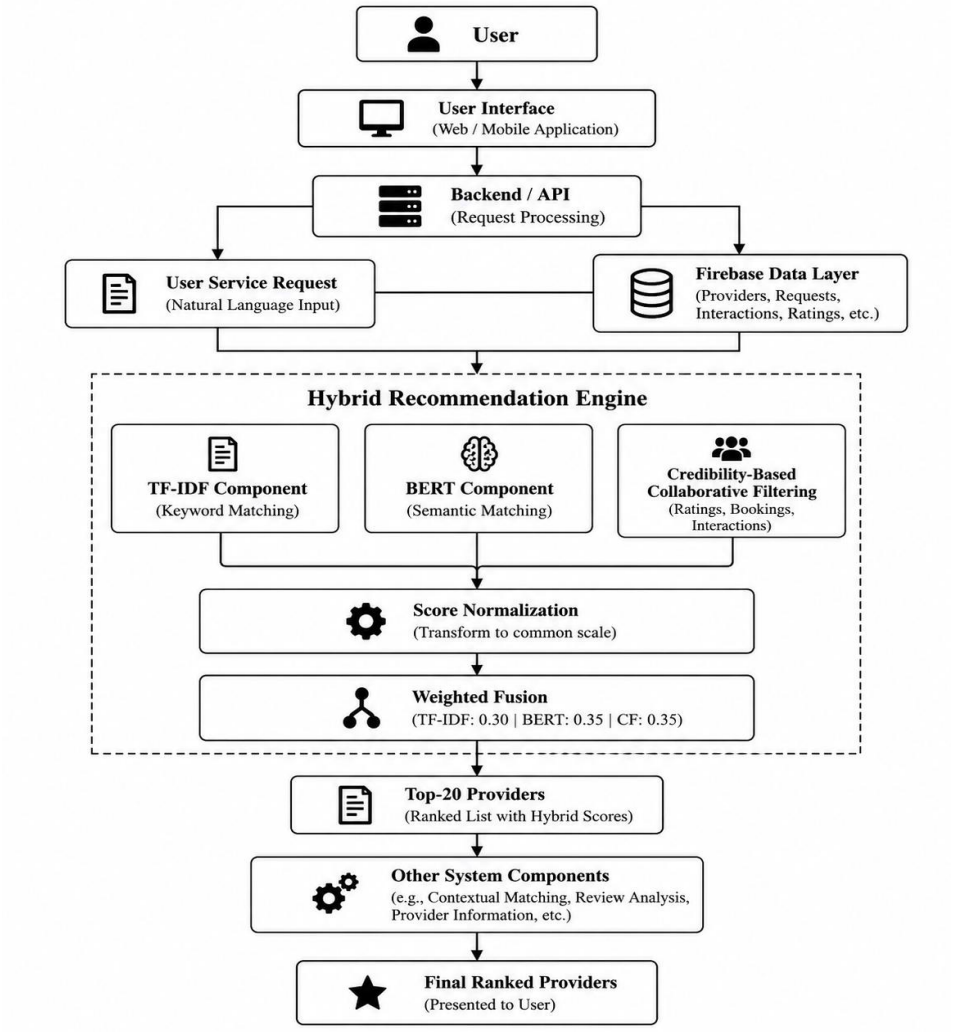


Figure 4.1:Architecture of the Hybrid Recommendation Component

4.3 Design alternatives and decisions

4.3.1 TF-IDF as a Standalone Approach

TF-IDF was considered because it is computationally efficient and effective for identifying direct keyword similarity between user requests and provider descriptions. However, a standalone TF-IDF solution has difficulty understanding requests where users and providers express the same requirement using different words. For example, semantically similar service requirements may contain limited exact keyword overlap. Therefore, TF-IDF was retained as a lexical relevance component rather than being used as the complete recommendation solution.

4.3.2 BERT as a Standalone Approach

A multilingual sentence-transformer model was considered to overcome the semantic limitations of traditional keyword matching. BERT-based embeddings can capture contextual meaning and identify semantic similarity between differently worded sentences.

However, semantic similarity alone does not represent provider credibility or previous user-provider interactions. Therefore, BERT was selected as the semantic component but was not used as the only ranking mechanism.

4.3.3 Traditional Collaborative Filtering [5], [6]

Collaborative filtering was considered for incorporating historical user-provider interactions and personalization. However, traditional interaction-based recommendation can perform poorly when historical interactions are limited. Furthermore, interaction history alone does not explicitly represent several provider-level quality indicators required in a service marketplace.

For this reason, a customized credibility-based collaborative filtering approach was designed to include provider rating, successful booking outcomes, interaction frequency, and previous user-provider interactions.

4.3.4 4.3.4 Selection of the Hybrid Architecture

Based on the limitations of the individual alternatives, a hybrid architecture was selected. The final design combines TF-IDF, multilingual BERT, and credibility-based collaborative filtering. The three components provide complementary information: TF-IDF represents lexical relevance, BERT represents semantic relevance, and the collaborative filtering component represents credibility and historical personalization.

Table 4.3: Comparison of Design Alternatives

Approach	Strength	Limitation	Design Decision
TF-IDF	Effective keyword matching and low	Limited semantic understanding	Included as lexical component

	complexity		
BERT	Strong contextual and semantic matching	Does not directly model provider credibility	Included as semantic component
Traditional CF	Uses historical interactions	Cold-start and limited explicit quality representation	Extended with credibility information
Hybrid Approach	Combines complementary recommendation signals	Higher implementation complexity	Selected as final design

4.4 Detailed design

4.4.1 Recommendation Processing Workflow

The recommendation process begins when a user submits a natural-language description of the required service. The request and candidate provider information are processed through TF-IDF and BERT to obtain lexical and semantic similarity scores. In parallel, provider credibility and available historical interaction information are processed by the collaborative filtering component.

Because the three recommendation techniques produce scores with different characteristics, the individual scores are normalized before fusion. The normalized values are combined using predefined weights, after which providers are sorted according to the final hybrid score. The highest-ranked 20 providers are selected as the output of the recommendation component.

4.4.2 TF-IDF Component Design

The TF-IDF component was designed to capture lexical similarity between a user request and provider descriptions. Provider textual information is transformed into TF-IDF representations, and the user request is represented within the same feature

space. Similarity between the request and each provider is then determined using cosine similarity.

This component provides a strong signal when the user request contains service-related terms that also appear in the provider description. It therefore acts as the keyword-level relevance component of the hybrid architecture.

4.4.2 Multilingual BERT Component Design

The BERT component was introduced to capture semantic relationships that may not be identified through exact keyword matching. The design uses the pretrained paraphrase-multilingual-MiniLM-L12-v2 sentence-transformer model to generate vector representations of user requests and provider descriptions.

Semantic similarity is obtained by comparing these representations. This allows providers to receive a high semantic relevance score even when their descriptions do not contain exactly the same words as the user request but communicate a similar meaning.

4.4.4 Credibility-Based Collaborative Filtering Design

The collaborative filtering component was customized to incorporate provider credibility in addition to historical user-provider interactions. Three provider-level indicators were selected: rating, booking success rate, and interaction frequency.

Rating was assigned the greatest importance because it directly represents customer-perceived service quality. Booking success rate represents provider reliability, while interaction frequency represents activity and experience. Interaction frequency was given a lower contribution because a high number of interactions does not necessarily indicate high service quality.

A transformation is applied to interaction frequency to reduce the effect of extremely active providers. Missing information is handled using a neutral value where required. In addition, previous interaction between a user and provider is used as a personalization signal, providing a controlled boost when prior history exists.

Table 4.4:Credibility Factors

Factor	Weight	Design Rationale
Provider Rating	0.50	Represents customer-perceived service quality
Booking Success Rate	0.30	Represents reliability in completing bookings
Interaction Frequency	0.20	Represents provider activity and experience

The resulting credibility information is combined with historical interaction information to produce the collaborative filtering score. This design allows the recommendation framework to consider provider quality while retaining a personalization signal from previous user-provider relationships.

4.4.5 Score Normalization and Hybrid Fusion

The individual recommendation components generate scores with different distributions. Directly combining these values could allow one component to dominate the final ranking because of its numerical scale. Therefore, the component scores are normalized to a common range between 0 and 1 before score fusion.

The final hybrid configuration assigns a weight of 0.30 to TF-IDF, 0.35 to BERT, and 0.35 to credibility-based collaborative filtering. These weights were selected after examining different weight configurations through sensitivity analysis. The selected configuration provides a balance between lexical relevance, semantic relevance, and credibility/personalization information.

Table 4.5:Final Hybrid Score Configuration

Component	Weight	Role
TF-IDF	0.30	Keyword/lexical relevance

Multilingual BERT	0.35	Semantic relevance
Credibility CF	0.35	Credibility and personalization

4.4.6 Provider Ranking and Output

After score fusion, candidate providers are sorted in descending order according to their final hybrid recommendation scores. The Top-20 providers are selected as the recommendation component output.

Each recommendation result can include the provider identifier and the relevant recommendation scores required by the remaining system components. This modular output design allows the recommendation engine to operate independently while supporting integration with the complete service provider matching platform.

4.5 Standards and professional practice

The development of the recommendation component followed software engineering and machine learning practices intended to improve maintainability, reliability, reproducibility, and responsible use.

The implementation was organized into separate logical stages for data preprocessing, TF-IDF matching, BERT-based semantic matching, credibility-based collaborative filtering, score normalization, hybrid fusion, and evaluation. This modular structure supports easier testing, debugging, and future modification of individual recommendation components.

Version control should be maintained using Git to track implementation changes and support collaborative development. Sensitive configuration information, including database connection credentials, should be separated from application source code and should not be committed to public repositories.

Consistent data preprocessing and evaluation procedures were maintained throughout the experiments. A fixed random state was used for the evaluation sampling process to improve reproducibility. The individual recommendation approaches were evaluated under the same evaluation conditions to support a fair comparison.

Professional practice also includes responsible handling of user and provider information. Only information required for recommendation generation should be

processed, and personally identifiable information should not be unnecessarily incorporated into machine learning calculations. Appropriate database authentication, access control, and secure credential management should be applied when deploying the solution.

The recommendation output should also be interpreted as decision-support information rather than an absolute judgement of service provider quality. Ratings and historical interactions can contain bias, while new providers may have limited historical information. Therefore, the modular hybrid design and future monitoring of recommendation behaviour are important for maintaining fairness and reliability.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The Hybrid Recommendation Framework was developed using Python and commonly used machine learning and data-processing libraries. Google Colab was used during model development and experimentation because it provides a convenient Python-based environment for executing data preprocessing, recommendation algorithms, and evaluation procedures.

Pandas and NumPy were used for dataset manipulation and numerical operations. Scikit-learn was used to implement TF-IDF vectorization, cosine similarity calculations, normalization, and other supporting machine learning operations. The Sentence Transformers library was used to implement semantic matching using the pretrained paraphrase-multilingual-MiniLM-L12-v2 model.

Firebase was used as the database and application data service. It supports the storage and retrieval of information required by the complete service provider matching platform, including provider-related information and application data used by the recommendation workflow.

The implementation was organized into separate logical modules so that individual recommendation methods could be independently developed, tested, and integrated.

Table 5.1:Development Environment and Technology Stack

Technology	**Purpose**
Python	Core recommendation algorithm implementation
Google Colab	Model development, experimentation and evaluation
Pandas	Dataset loading, cleaning and manipulation
NumPy	Numerical processing
Scikit-learn	TF-IDF, cosine similarity and preprocessing

Sentence Transformers	Multilingual BERT sentence embeddings
paraphrase-multilingual-MiniLM-L12-v2	Semantic similarity model
Firebase	Application database and data management
Git / GitHub	Version control and source-code management

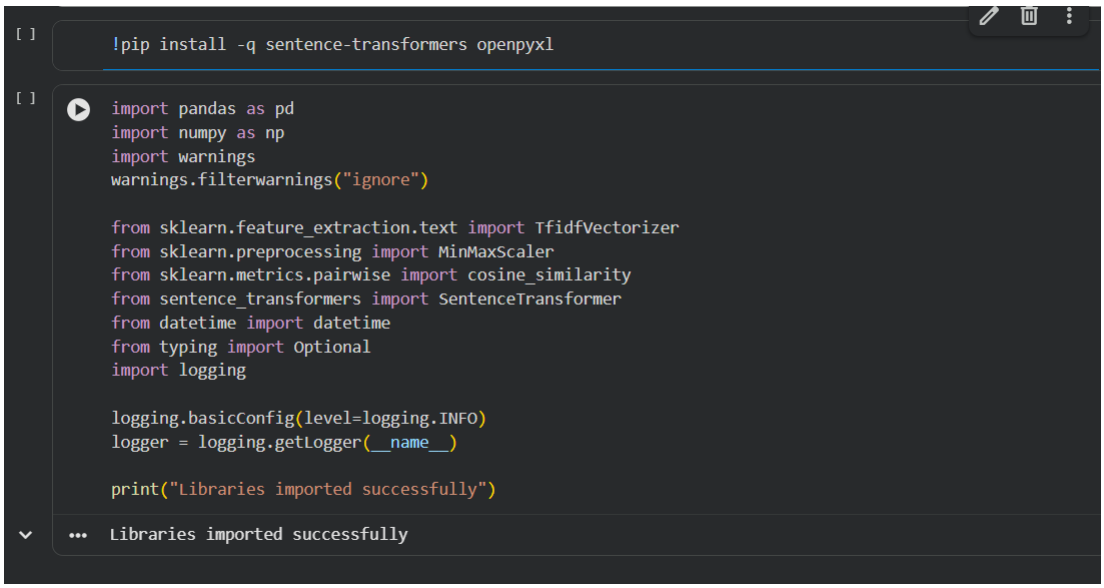


Figure 5.1:Development Environment Used for Recommendation Model Implementation

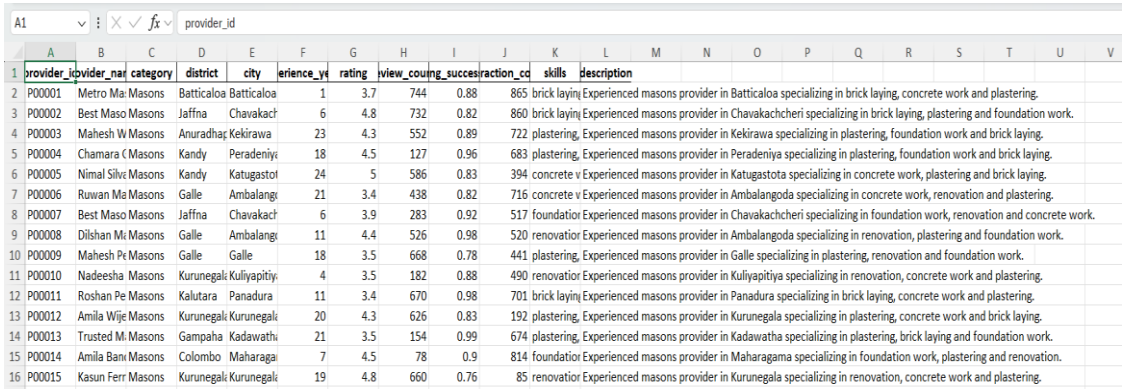
5.2 Component implementation

5.2.1 Data Preparation and Preprocessing

Before executing the recommendation algorithms, provider, user request, and interaction-related data were prepared for processing. The preprocessing stage was responsible for checking the required attributes, handling missing values where necessary, and preparing textual and numerical information for the recommendation algorithms.

Provider service descriptions were prepared for TF-IDF and semantic similarity processing. User service requests were used as recommendation queries, while historical interaction and provider-related information were prepared for the credibility-based collaborative filtering component.

Maintaining a common preprocessing pipeline ensured that the recommendation components received data in a consistent format.



provider_id	provider_nar	category	district	city	erience_ye	rating	view_cou	ng_succes	raction_co	skills	description
P00001	Metro Ma	Masons	Batticaloa	Batticaloa	1	3.7	744	0.88	865	brick layin	Experienced masons provider in Batticaloa specializing in brick laying, concrete work and plastering.
P00002	Best Maso	Masons	Jaffna	Chavakach	6	4.8	732	0.82	860	brick layin	Experienced masons provider in Chavakachcheri specializing in brick laying, plastering and foundation work.
P00003	Mahesh W	Masons	Anuradhap	Kekirawa	23	4.3	552	0.89	722	plastering,	Experienced masons provider in Kekirawa specializing in plastering, foundation work and brick laying.
P00004	Chamara (	Masons	Kandy	Peradeniya	18	4.5	127	0.96	683	plastering,	Experienced masons provider in Peradeniya specializing in plastering, foundation work and brick laying.
P00005	Nimal Silva	Masons	Kandy	Katugastot	24	5	586	0.83	394	concrete v	Experienced masons provider in Katugastota specializing in concrete work, plastering and brick laying.
P00006	Ruwan Ma	Masons	Galle	Ambalang	21	3.4	438	0.82	716	concrete v	Experienced masons provider in Ambalangoda specializing in concrete work, renovation and plastering.
P00007	Best Maso	Masons	Jaffna	Chavakach	6	3.9	283	0.92	517	foundatior	Experienced masons provider in Chavakachcheri specializing in foundation work, renovation and concrete work.
P00008	Dilshan Ma	Masons	Galle	Ambalang	11	4.4	526	0.98	520	renovatior	Experienced masons provider in Ambalangoda specializing in renovation, plastering and foundation work.
P00009	Mahesh Pe	Masons	Galle	Galle	18	3.5	668	0.78	441	plastering,	Experienced masons provider in Galle specializing in plastering, renovation and foundation work.
P00010	Nadeesha	Masons	Kurunegal	Kuliyapitiy	4	3.5	182	0.88	490	renovatior	Experienced masons provider in Kuliyapitiya specializing in renovation, concrete work and plastering.
P00011	Roshan Pe	Masons	Kalutara	Panadura	11	3.4	670	0.98	701	brick layin	Experienced masons provider in Panadura specializing in brick laying, concrete work and plastering.
P00012	Amila Wije	Masons	Kurunegal	Kurunegal	20	4.3	626	0.83	192	plastering,	Experienced masons provider in Kurunegala specializing in plastering, concrete work and brick laying.
P00013	Trusted M	Masons	Gampaha	Kadawath	21	3.5	154	0.99	674	plastering,	Experienced masons provider in Kadawatha specializing in plastering, brick laying and foundation work.
P00014	Amila Ban	Masons	Colombo	Maharaga	7	4.5	78	0.9	814	foundatior	Experienced masons provider in Maharagama specializing in foundation work, plastering and renovation.
P00015	Kasun Ferr	Masons	Kurunegal	Kurunegal	19	4.8	660	0.76	85	renovatior	Experienced masons provider in Kurunegala specializing in renovation, concrete work and plastering.

Figure 5.2:Sample of Preprocessed Provider Data

5.2.2 TF-IDF Implementation

The TF-IDF component was implemented using the TF-IDF vectorization facilities provided by Scikit-learn. Provider service descriptions were transformed into numerical TF-IDF vectors representing the importance of words within the available textual information.

When a user submits a service request, the request is transformed using the same TF-IDF representation. Cosine similarity is then used to compare the request vector against provider vectors. This produces a TF-IDF similarity score for each candidate provider.

The generated score represents lexical similarity and is particularly useful when the user's request contains keywords that directly correspond to the services described by a provider.

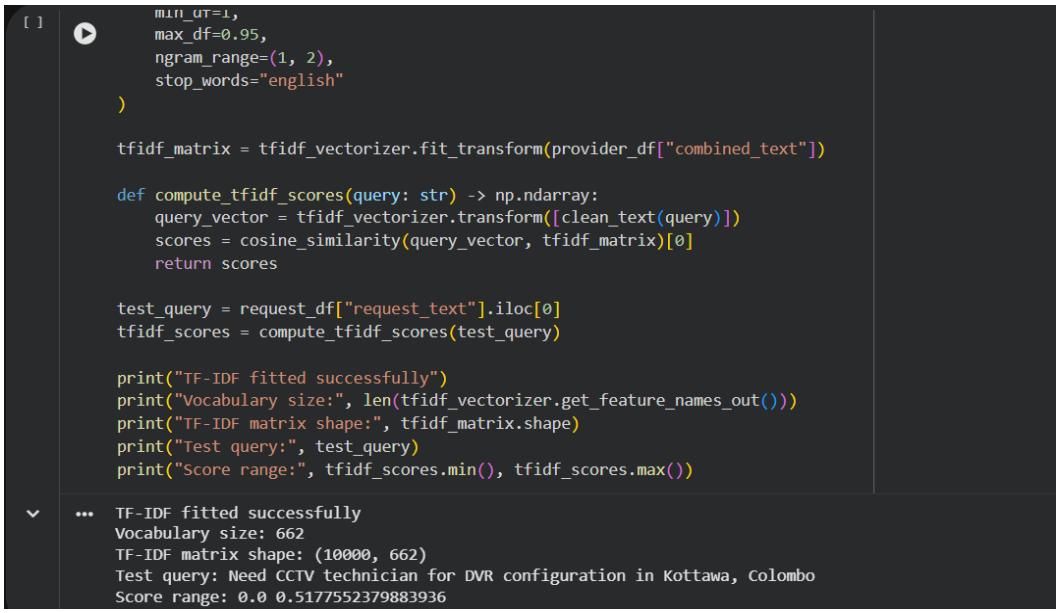


Figure 5.3:TF-IDF output

5.2.3 Multilingual BERT Implementation

Semantic matching was implemented using the pretrained paraphrase-multilingual-MiniLM-L12-v2 model through the Sentence Transformers library.

The model transforms both user requests and provider descriptions into dense sentence embeddings. Similarity between the request embedding and provider embeddings is then calculated to obtain a semantic relevance score.

Unlike TF-IDF, this component is not limited to direct keyword overlap. Consequently, semantically related requests and provider descriptions can obtain strong similarity scores even when different words are used to express the service requirement.

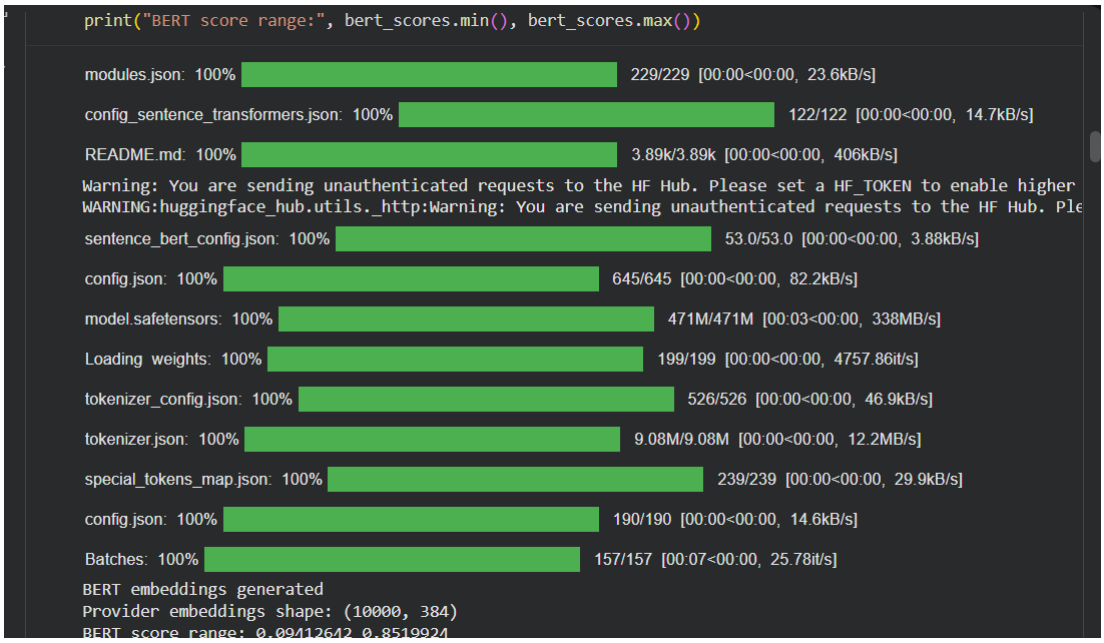


Figure 5.4:BERT based semantic matching

5.2.4 Credibility-Based Collaborative Filtering Implementation

The collaborative filtering component was customized to incorporate provider credibility together with historical interaction information. Provider credibility was represented using three indicators: provider rating, successful booking rate, and interaction frequency.

The implementation assigns a weight of 0.50 to provider rating, 0.30 to booking success rate, and 0.20 to interaction frequency. Interaction frequency is transformed to reduce the possibility of highly active providers dominating the credibility score solely because of their number of interactions.

Where appropriate, missing values are handled using a neutral value. Previous interaction between the current user and a provider is also incorporated as a personalization signal. Providers with relevant previous user-provider interaction can receive a controlled history-based adjustment. The resulting value is constrained to the normalized scoring range before being passed to the hybrid recommendation stage.

```
        if user_id in user_preferences:
            user_history = user_preferences[user_id]

            for prov_id in user_history:
                provider_idx = provider_df.index[provider_df["provider_id"] == prov_id].tolist()

                if len(provider_idx) > 0:
                    cf_scores[provider_idx[0]] *= 1.20

        cf_scores = np.clip(cf_scores, 0, 1)
        cf_scores = np.nan_to_num(cf_scores, nan=0.5)
        return cf_scores

    test_user_id = interaction_df["user_id"].iloc[0]
    cf_scores = compute_cf_scores(test_user_id)

    print("Custom CF scores computed")
    print("Test user:", test_user_id)
    print("CF score range:", cf_scores.min(), cf_scores.max())
    print("Mean CF score:", cf_scores.mean())
```

```
Custom CF scores computed
Test user: U03894
CF score range: 0.0 0.9999999999999999
Mean CF score: 0.5572621570720052
```

Figure 5.5:Sample output of CF

5.2.5 Score Normalization and Hybrid Fusion

The three recommendation components generate scores with different statistical characteristics. Therefore, the generated TF-IDF, BERT, and collaborative filtering scores are normalized to a common range before they are combined.

The final implementation uses the hybrid configuration identified during model development: 0.30 for TF-IDF, 0.35 for BERT, and 0.35 for credibility-based collaborative filtering.

After score fusion, candidate providers are sorted according to their final hybrid scores. The providers with the highest scores are selected as the Top-20 recommendation candidates.

```
top20.head(10)
```

review_count	booking_success_rate	interaction_count	hybrid_score	tfidf_score	bert_score	cf_score
65	0.444444	9.0	0.900299	0.983842	0.976433	0.752558
667	0.600000	5.0	0.895471	0.958703	0.947832	0.788910
257	0.600000	10.0	0.892001	0.980593	0.959490	0.748576
448	0.357143	14.0	0.889186	0.980760	0.989811	0.710070

Figure 5.6:Sample Output of Hybrid Score Generation

5.3 Integration with the other components

The Hybrid Recommendation Framework was integrated as one intelligent component of the complete service provider matching platform. The recommendation component receives the user's service requirement and obtains the information required for provider matching through the application data layer.

Firebase is used to support data storage and retrieval within the application. Provider-related information and other required application records can therefore be accessed by the relevant system services during the recommendation workflow.

The recommendation component processes candidate providers using TF-IDF, multilingual BERT, and credibility-based collaborative filtering. After normalization and score fusion, a ranked list of the Top-20 provider candidates is generated.

The ranked candidate list is then made available to the other intelligent components of the complete system. These components can apply additional information such as temporal and contextual suitability, provider-related verification information, and review-related analysis. The complete platform can then generate the final provider recommendations presented through the user interface.

5.4 Verification and software testing

5.4.1 Unit and Functional Testing

Individual recommendation functions were tested separately. TF-IDF testing verified that valid user requests generated lexical similarity scores. BERT testing verified the generation of semantic similarity scores, while collaborative filtering testing examined credibility calculation and handling of available historical interaction information.

The normalization and hybrid fusion stages were also tested to ensure that the component scores were correctly combined and that candidate providers were sorted in descending order according to the final hybrid score.

Table 5.2:Functional test cases

Test ID	Test Scenario	Expected Result	Actual Result
TC01	Submit valid service request	Request is accepted for recommendation processing	Pass
TC02	Execute TF-IDF matching	TF-IDF similarity scores are generated	Pass
TC03	Execute BERT matching	Semantic similarity scores are generated	Pass
TC04	Calculate provider credibility	Valid credibility/CF scores are generated	Pass
TC05	Missing historical information	System handles missing information without pipeline failure	Pass
TC06	Normalize component scores	Scores are transformed to expected common range	Pass
TC07	Generate hybrid score	TF-IDF, BERT and CF scores are successfully	Pass

		combined	
TC08	Rank providers	Providers are ordered by final hybrid score	Pass
TC09	Generate Top-20	Maximum required recommendation candidates are returned	Pass
TC10	Retrieve application data	Required records are successfully accessed from Firebase	Pass

5.4.2 Recommendation Performance Testing

In addition to functional testing, recommendation quality was evaluated using Precision@K and NDCG@K at K=5, K=10 and K=20. The same evaluation procedure was applied to TF-IDF, BERT, collaborative filtering, and the proposed hybrid method to enable a consistent comparison.

A fixed evaluation sample was used to improve experimental reproducibility. Category-level relevance was used as the ground-truth criterion, where a provider was treated as relevant when the provider service category matched the category associated with the corresponding user request.

The detailed experimental results are presented and analyzed in Chapter 6.



Figure 5.7:Integrated Recommendation Output Generated by the System

5.5 Deployment and operation

The implemented recommendation framework was designed to operate as part of the complete service provider matching application. Firebase provides the application data layer, while the recommendation logic is executed through the backend/model-processing environment.

During operation, a user submits a service requirement through the user interface. The backend processes the request and obtains the information required by the recommendation engine. The recommendation pipeline generates lexical, semantic, and credibility-based scores, combines the scores, and returns the Top-20 candidate providers. The candidate list is then made available to the remaining components of the system before the final recommendations are displayed.

The modular implementation supports future deployment improvements because individual recommendation components can be updated without redesigning the complete platform. Future operational improvements may include caching provider embeddings, optimizing model loading, monitoring response times, and updating provider information as new application data becomes available.

Security considerations are also important during deployment. Firebase access should be protected using appropriate authentication and security rules. Sensitive credentials and configuration values should not be exposed within public source-code repositories or client-side code. Application access should follow the minimum permissions required for each system component.

6 RESULTS AND VALIDATION

6.1 Research results

The main research evaluation compared the performance of the three individual recommendation components with the proposed hybrid approach. The same evaluation procedure was applied to all approaches to provide a consistent comparison.

A total of 500 user requests were randomly selected for evaluation using a fixed random seed. The service category associated with each request was used as the relevance ground truth. A recommended provider was considered relevant when the provider's service category matched the category associated with the corresponding user request.

Precision@K was used to measure the proportion of relevant providers appearing within the Top-K recommendations, while NDCG@K was used to measure ranking quality by considering the positions of relevant providers in the ranked list.

6.1.1 Recommendation Performance

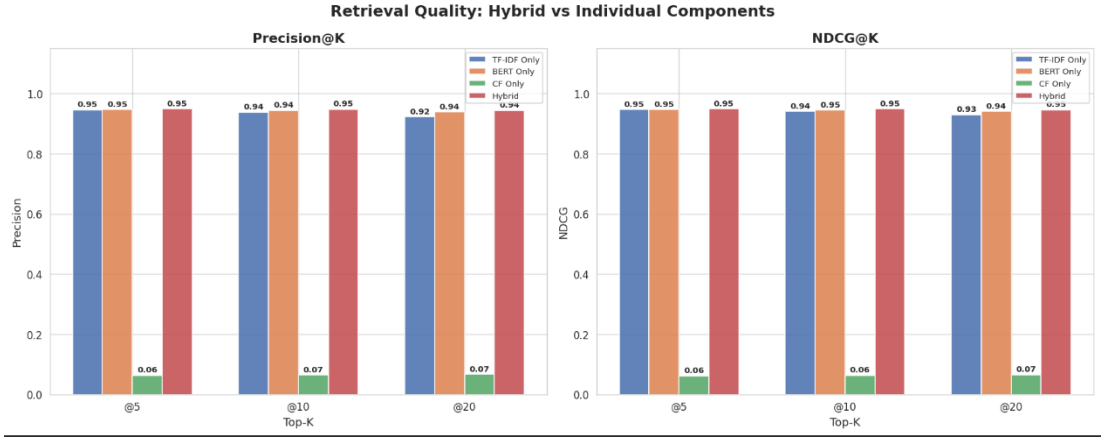


Figure 6.1:Recommendation Performance Comparison

The results demonstrate that the proposed Hybrid Recommendation Framework achieved the highest overall performance across the evaluated recommendation list sizes. At K=5, the Hybrid approach achieved a Precision value of 0.9490 and an NDCG value of 0.9496. At K=20, it maintained a Precision value of 0.9442 and an NDCG value of 0.9458.

TF-IDF and BERT individually achieved strong category-level relevance performance. BERT generally performed better than TF-IDF as the recommendation list size increased, demonstrating the value of semantic similarity in identifying providers whose descriptions are contextually related to user requests.

The collaborative filtering component produced considerably lower standalone Precision and NDCG values. This indicates that historical interaction and credibility information alone was not sufficient to determine service-category relevance. However, collaborative filtering was designed to provide a complementary credibility and personalization signal rather than functioning as the primary content-matching mechanism.

When the three signals were combined, the Hybrid approach produced the strongest overall performance. This supports the central design decision of combining lexical relevance, semantic understanding, and credibility-related information instead of relying on a single recommendation technique.

6.1.2 Score Distribution Analysis

The distribution of the scores generated by the individual recommendation components was analysed to understand their behaviour before hybrid fusion. TF-IDF, BERT, and collaborative filtering represent different recommendation signals and therefore produce different score distributions.

TF-IDF primarily represents direct lexical similarity, whereas BERT captures semantic relationships between requests and provider descriptions. The collaborative filtering component represents a different dimension based on credibility and historical interaction information. The differences observed among these score distributions further justify the normalization stage used before weighted score fusion.

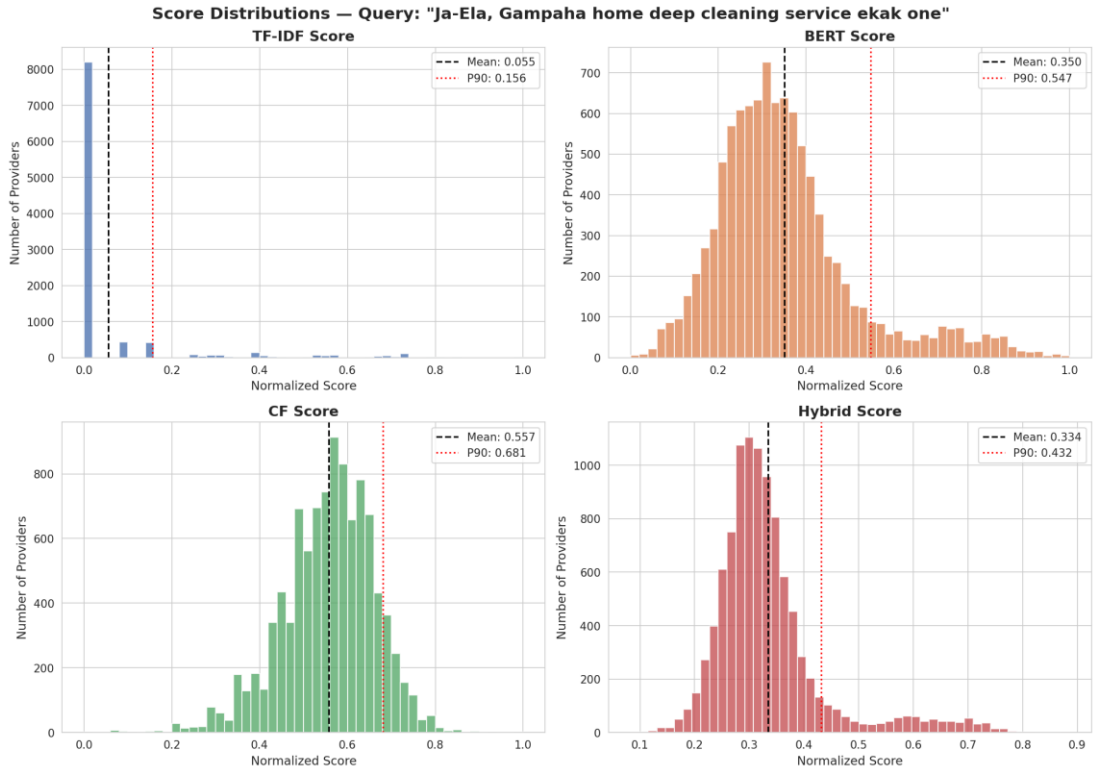


Figure 6.2:Distribution of TF-IDF, BERT and Collaborative Filtering Scores

6.1.3 Component Correlation Analysis

Correlation analysis was performed to examine the relationship between the scores produced by the individual recommendation components. The purpose of this analysis was to determine whether the components provided identical information or complementary recommendation signals.

TF-IDF and BERT both analyse the relationship between the service request and provider information, but they represent this relationship differently. TF-IDF focuses on lexical overlap, while BERT captures contextual and semantic similarity. The collaborative filtering component is based on credibility and historical interaction information and therefore provides a different type of recommendation signal.

The correlation analysis supports the use of multiple recommendation components because the individual methods contribute different information to the final ranking.

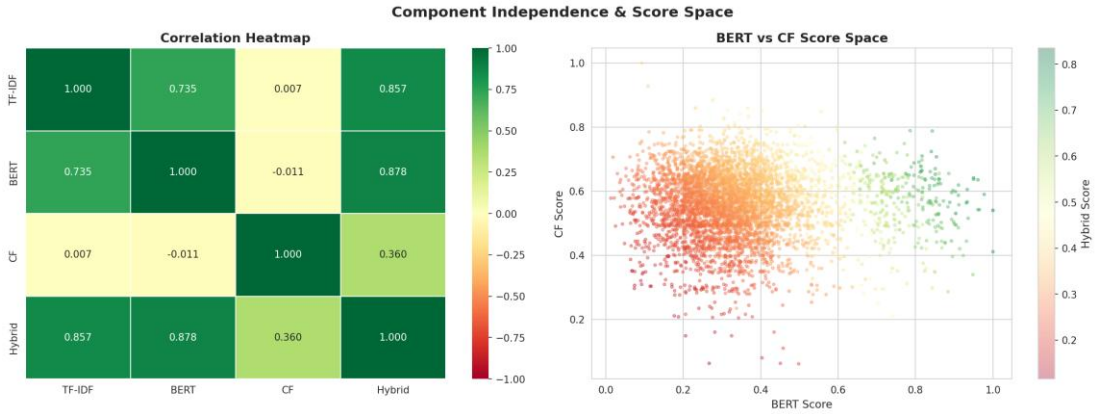


Figure 6.3:Pearson Correlation between Recommendation Component Scores

6.1.4 Hybrid Weight Sensitivity Analysis

A weight sensitivity analysis was conducted to examine how different combinations of component weights affected the performance of the hybrid recommendation model. This analysis was important because the final ranking depends on the relative contribution assigned to lexical relevance, semantic relevance, and collaborative filtering.

Different configurations were examined by varying the contribution of the BERT component while proportionally adjusting the remaining component weights. The final configuration of 0.30 for TF-IDF, 0.35 for BERT, and 0.35 for collaborative filtering provided stable performance for Precision@20 and NDCG@20 and was therefore selected for the final Hybrid Recommendation Framework.

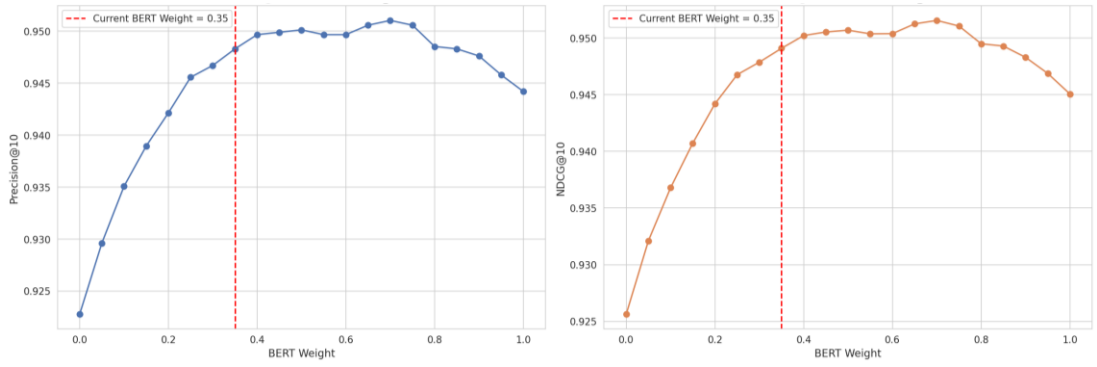


Figure 6.4:Effect of Hybrid Weight Configurations

6.2 Product and technical results

In addition to experimental recommendation performance, the implemented component was evaluated from a technical and product perspective. The complete recommendation pipeline was successfully implemented to accept a user service request, process provider information through the three recommendation components, normalize the generated scores, calculate the final hybrid score, and produce a ranked provider list.

The implementation demonstrated that TF-IDF and BERT could operate together to provide complementary textual matching capabilities. The credibility-based collaborative filtering component successfully introduced provider credibility and available historical interaction information into the ranking process.

The final output contains ranked provider candidates together with their calculated recommendation scores. The component is capable of producing the Top-20 provider candidates required by the subsequent stages of the complete service provider matching platform.

Firebase was used as the application data layer to support the storage and retrieval of information required by the platform. The modular structure of the recommendation component allows the recommendation logic to remain separated from the user interface and other intelligent components.

▼ View actual Component 1 Top-20 providers and selection reasons

Rank	Provider	Category / location	Hybrid	TF-IDF	Semantic	Preference	Why selected
#1	**Ajith Fernando** P05841	Cleaners Hikkaduwa, Galle	0.648	0.500	0.801	0.621	Selected through the exact category, district and city match tier with hybrid score 0.6477. Normalized signals: TF-IDF 0.5000, semantic 0.8013, collaborative preference 0.6208; strongest signal was semantic relevance.
#2	**Lanka Cleaners Maintenance** P05976	Cleaners Hikkaduwa, Galle	0.614	0.500	0.873	0.453	Selected through the exact category, district and city match tier with hybrid score 0.6139. Normalized signals: TF-IDF 0.5000, semantic 0.8728, collaborative preference 0.4527; strongest signal was semantic relevance.
#3	**Smart Cleaners Experts** P05760	Cleaners Hikkaduwa, Galle	0.612	0.500	0.852	0.467	Selected through the exact category, district and city match tier with hybrid score 0.6118. Normalized signals: TF-IDF 0.5000, semantic 0.8520, collaborative preference 0.4674; strongest signal was semantic relevance.
#4	**Sunil Karunaratne** P05805	Cleaners Galle, Galle	0.637	0.500	0.661	0.731	Selected through the category and district match tier with hybrid score 0.6372. Normalized signals: TF-IDF 0.5000, semantic 0.6613, collaborative preference 0.7306; strongest signal was collaborative preference.
#5	**Sanjeewa Gunawardena** P06011	Cleaners Ambalangoda, Galle	0.623	0.500	0.704	0.646	Selected through the category and district match tier with hybrid score 0.6226. Normalized signals: TF-IDF 0.5000, semantic 0.7044, collaborative preference 0.6459; strongest signal was semantic relevance.
#6	**Chamara Madushan** P06063	Cleaners Galle, Galle	0.616	0.500	0.674	0.658	Selected through the category and district match tier with hybrid score 0.6162. Normalized signals: TF-IDF 0.5000, semantic 0.6737, collaborative preference 0.6583; strongest signal was semantic relevance.
#7	**Nimal Perera** P05786	Cleaners Ambalangoda, Galle	0.607	0.500	0.737	0.569	Selected through the category and district match tier with hybrid score 0.6071. Normalized signals: TF-IDF 0.5000, semantic 0.7368, collaborative preference 0.5692; strongest signal was semantic relevance.

Figure 6.5:Ranked Provider Output

6.3 User or stakeholder validation

The primary validation of this research focused on technical and experimental evaluation of the recommendation framework. Recommendation quality was quantitatively evaluated using Precision@K and NDCG@K, while functional and integration testing was used to verify the technical behaviour of the implemented component.

A formal large-scale end-user study was outside the scope of the current research. Therefore, the reported results should primarily be interpreted as evidence of recommendation relevance and technical feasibility rather than direct evidence of user satisfaction.

The developed prototype nevertheless demonstrates the intended user workflow in which a service request is submitted and a ranked set of provider candidates is generated. Future work should extend the validation through structured user studies involving service customers and providers. Such evaluation could measure perceived recommendation relevance, ease of use, trust in provider rankings, and overall satisfaction with the recommendation process.

6.4 Integration results

The Hybrid Recommendation Framework was developed as one component of the complete intelligent service provider matching platform. Integration testing focused on verifying whether the recommendation component could receive the required input, generate ranked provider candidates, and provide its output in a form suitable for subsequent system components.

The integrated workflow begins with the user service request. The request is transferred through the application layer to the recommendation process, where provider information is analysed using TF-IDF, multilingual BERT, and credibility-based collaborative filtering. The generated component scores are normalized and fused to produce the Top-20 provider candidates.

The resulting candidate list can then be processed by the other intelligent modules of the complete platform, including contextual and temporal matching, provider-related verification, and review-related analysis. This modular integration allows each research component to contribute a different type of intelligence to the final service provider matching process.

Integration testing also demonstrated the benefit of maintaining a standardized provider representation across the components. Provider identifiers and required recommendation information can be transferred between modules without tightly coupling the internal implementation of the individual machine learning components.

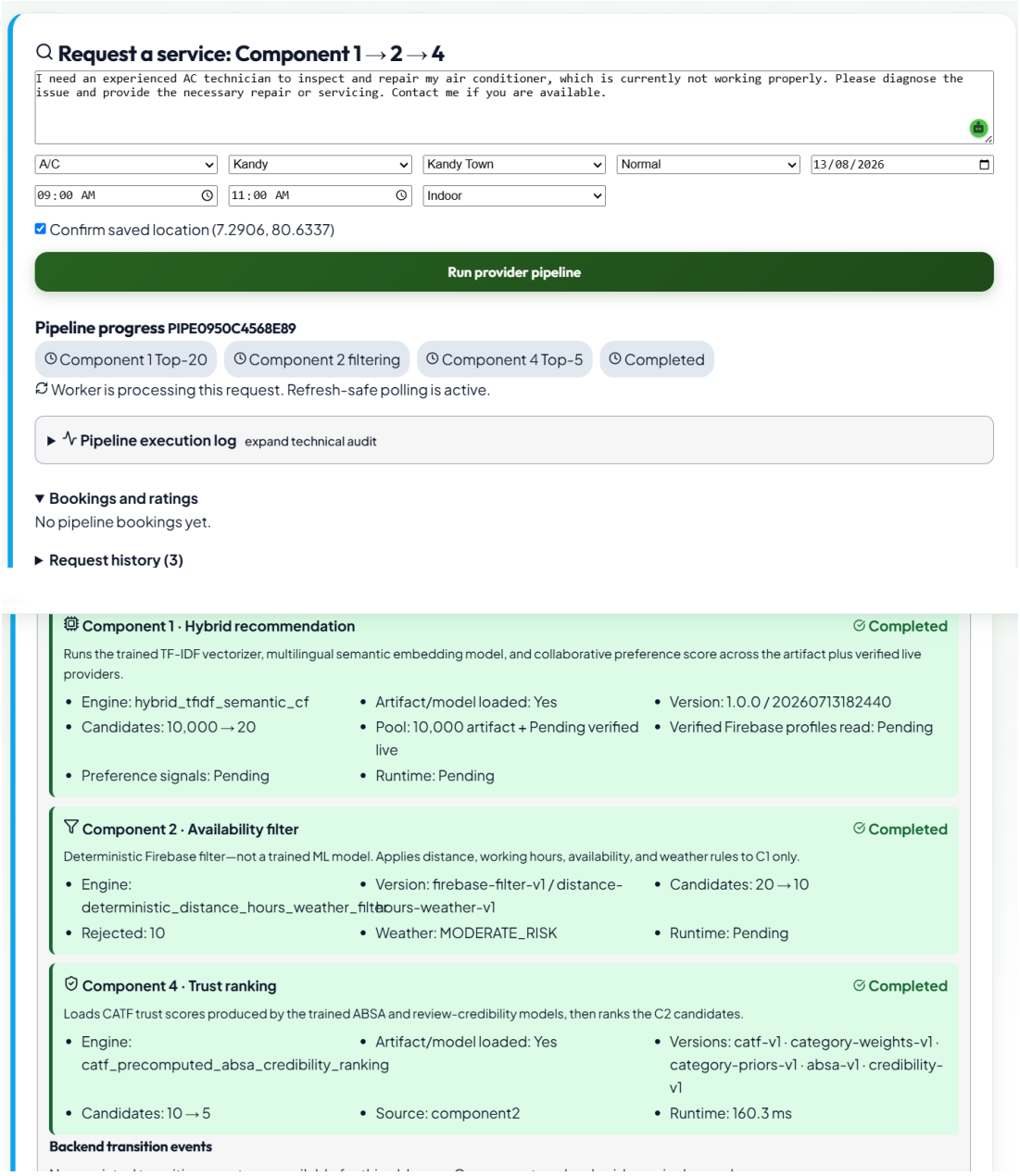


Figure 6.6:Integrated Service Provider Recommendation Output Presented through the Application

6.5 Summary of findings

The experimental and technical evaluation produced several important findings regarding the proposed Hybrid Recommendation Framework.

First, both TF-IDF and multilingual BERT demonstrated strong performance in identifying service providers relevant to user requests. TF-IDF provided effective

lexical matching, while BERT provided semantic matching capabilities that were less dependent on exact keyword overlap.

Second, the credibility-based collaborative filtering component achieved substantially lower category-level Precision and NDCG when evaluated independently. This result indicates that credibility and historical interaction information should not be used as the only mechanism for identifying service-category relevance. Instead, it is more suitable as a complementary signal that introduces provider credibility and personalization information into the recommendation process.

Third, combining the three recommendation signals produced the strongest overall results. The proposed Hybrid approach achieved a Precision@5 of 0.9490 and an NDCG@5 of 0.9496, while maintaining strong performance at larger recommendation list sizes. The Hybrid method also achieved a Precision@20 of 0.9442 and an NDCG@20 of 0.9458**.**

Fourth, score distribution and correlation analyses demonstrated that the individual components exhibit different behaviours and contribute different types of information. The use of score normalization therefore provided a common basis for hybrid fusion.

Finally, weight sensitivity analysis supported the selected hybrid configuration of 0.30 TF-IDF, 0.35 BERT, and 0.35 credibility-based collaborative filtering. The results demonstrate that combining lexical, semantic, and credibility-related information provides a more balanced recommendation strategy than relying on an individual recommendation method.

However, the evaluation uses service-category matching as the primary relevance ground truth**.** Therefore, the reported Precision and NDCG values primarily demonstrate category-level recommendation relevance and ranking quality. They should not be interpreted as direct measurements of real-world customer satisfaction or guaranteed provider quality. Future validation with real users and real-world service outcomes would provide stronger evidence of practical recommendation effectiveness.

7 DISCUSSION

7.1 Interpretation of findings

The experimental findings demonstrate that the proposed Hybrid Recommendation Framework achieved stronger overall recommendation performance than the individual TF-IDF, BERT, and credibility-based collaborative filtering approaches under the selected evaluation protocol. The hybrid method achieved a Precision@5 of 0.9490 and an NDCG@5 of 0.9496. At a larger recommendation list size of 20, the hybrid approach maintained a Precision@20 of 0.9442 and an NDCG@20 of 0.9458. These results indicate that the framework was effective in placing service-category-relevant providers within the highest-ranked recommendations.

The strong performance of TF-IDF demonstrates that lexical matching remains useful for service provider recommendation. Service requests frequently contain specific service-related terms, such as electrical repair, plumbing, or air-conditioning maintenance, which may also appear in provider descriptions. Therefore, direct keyword matching can identify many relevant providers without requiring complex semantic processing. However, lexical matching is less effective when users describe their requirements using different expressions from those used in provider profiles.

The multilingual BERT component addressed this limitation by representing service requests and provider descriptions through semantic embeddings. The BERT-only approach achieved a Precision@20 of 0.9387, compared with 0.9231 for TF-IDF. This difference suggests that semantic similarity provides additional value when the recommendation list includes a larger number of providers. By identifying contextual relationships rather than depending entirely on shared keywords, semantic matching can improve the identification of relevant service providers.

In contrast, the credibility-based collaborative filtering component achieved considerably lower standalone performance. Its Precision@5 was 0.0633, while its NDCG@5 was 0.0613. This result is important because the collaborative filtering component evaluates provider credibility and historical interactions rather than primarily matching service descriptions with user requirements. A provider may have strong ratings, successful bookings, and frequent interactions while belonging to a

different service category from the current request. Therefore, credibility information alone is insufficient for determining service-category relevance.

The hybrid framework was designed to address these differences by combining complementary recommendation signals. TF-IDF contributes lexical relevance, BERT contributes semantic relevance, and credibility-based collaborative filtering contributes information about provider quality indicators and previous interactions. The improved hybrid results suggest that these signals can be combined to produce a more effective ranking under the category-based evaluation protocol.

Nevertheless, the improvement over the strongest individual content-matching method was relatively small. At K=20, the hybrid method achieved a Precision improvement of 0.0055 over BERT, equivalent to 0.55 percentage points. This indicates an incremental rather than a dramatic improvement in category-level relevance. The practical importance of the credibility component may therefore be more apparent in future evaluations that directly measure provider reliability, booking outcomes, and user satisfaction.

The findings also demonstrate the importance of evaluating recommendation components according to their intended purposes. A component designed to represent credibility should not be judged solely by its ability to identify the correct service category. Consequently, the low standalone collaborative filtering results do not necessarily establish that the credibility calculation is ineffective. Instead, they indicate that the current relevance metric does not fully capture the contribution that credibility-related information is intended to provide.

7.2 Comparison with literature baselines and existing solutions

Existing recommendation research has introduced several approaches for identifying relevant items or services based on content, user preferences, and historical interactions. Content-based recommendation techniques commonly rely on textual features to identify similarities between user requirements and available items. Collaborative filtering approaches, on the other hand, use historical user-item relationships to estimate preferences and generate personalized recommendations.

More recent semantic recommendation methods use contextual language representations to capture meaning beyond direct keyword overlap.

The TF-IDF component of the proposed framework follows established information retrieval principles by representing textual information according to term importance and comparing documents through vector similarity. This approach provides an interpretable and computationally practical mechanism for matching service requests with provider descriptions. However, as discussed in information retrieval literature, term-based methods do not inherently understand the contextual meaning of language.

Transformer-based language models, including BERT and Sentence-BERT, provide an alternative by learning contextual representations of text. The multilingual sentence-transformer used in this research extends the matching process beyond exact word overlap. The experimental findings are consistent with the general motivation for semantic text matching, as the BERT component demonstrated strong recommendation performance, particularly at larger ranking cut-offs.

Traditional collaborative filtering methods, including neighbourhood-based techniques and matrix factorization approaches, primarily use patterns within historical interactions to identify user preferences. Although these methods are useful in domains with extensive interaction histories, service provider matching introduces additional challenges. Provider selection depends not only on previous preferences but also on service relevance, provider reliability, and successful completion of service requests. [5], [6]

The proposed research addresses this challenge through a customized credibility-based collaborative filtering component. Instead of depending exclusively on historical user-provider interaction patterns, the component incorporates provider rating, booking success rate, interaction frequency, and available user-provider history. This introduces service-related credibility indicators into the recommendation process.

The proposed hybrid framework differs from the individual baseline approaches by combining lexical, semantic, and credibility-related information within a single

weighted ranking process. This design is particularly relevant to service marketplaces, where the most textually relevant provider may not necessarily be the most reliable provider, and a highly rated provider may not necessarily offer the service required by the current user.

7.3 Research contribution

The primary research contribution of this study is the design and evaluation of a hybrid recommendation framework that integrates lexical similarity, multilingual semantic similarity, and provider credibility information for intelligent service provider matching.

The first contribution is the development of a unified recommendation architecture that combines TF-IDF, multilingual BERT, and credibility-based collaborative filtering. Each component addresses a different aspect of the recommendation problem. TF-IDF identifies direct textual relevance, BERT captures semantic relationships, and collaborative filtering introduces historical and credibility-related information. Combining these signals allows the framework to consider multiple dimensions of provider matching rather than depending on a single recommendation method.

The second and most significant contribution is the customization of the collaborative filtering component to incorporate provider credibility indicators. The proposed approach considers provider rating, booking success rate, and interaction frequency as separate indicators of service quality, reliability, and activity. These indicators are combined using domain-informed weights of 0.50, 0.30, and 0.20, respectively.

Interaction frequency is transformed to reduce the influence of providers with exceptionally high interaction counts. This design decision helps prevent provider activity alone from dominating the credibility score. Historical user-provider interactions are also considered to introduce a personalization element when relevant previous interaction information is available.

The third contribution is the integration of the three recommendation components through score normalization and weighted fusion. Because the individual

components generate different types of scores, normalization provides a common numerical basis for combining their outputs. The selected hybrid configuration assigns weights of 0.30 to TF-IDF, 0.35 to BERT, and 0.35 to credibility-based collaborative filtering. Weight sensitivity analysis was used to examine the behaviour of the framework under different component weight configurations.

The fourth contribution is the experimental comparison of the individual recommendation approaches and the proposed hybrid framework. The evaluation used Precision@K and NDCG@K at K values of 5, 10, and 20, allowing the recommendation methods to be examined across different ranking depths. The results provide evidence that the hybrid framework improves category-level recommendation relevance compared with the individual methods under the defined evaluation protocol.

Finally, the research contributes a modular recommendation design that can be extended with additional intelligent matching components. The separation of lexical, semantic, and credibility-related processing supports future improvements without requiring a complete redesign of the recommendation framework.

It is important to distinguish the proposed contribution from the invention of a completely new machine learning algorithm. TF-IDF, BERT, and collaborative filtering are established techniques. The contribution of this research lies in their application-specific integration, the customized credibility-based scoring approach, and the evaluation of their combined performance for service provider matching.

7.4 Product contribution

In addition to its research contribution, the proposed Hybrid Recommendation Framework provides a practical software component for an intelligent service provider matching platform. The component transforms a natural-language service request into a ranked list of candidate service providers, reducing dependence on manual provider searching and comparison.

A major product contribution is the ability to support natural-language service requests. Users are not required to express their requirements exclusively through predefined keywords. The combination of TF-IDF and multilingual BERT allows the

recommendation engine to process both direct service terminology and semantically related descriptions. This provides a more flexible approach to interpreting user requirements.

Another product contribution is the inclusion of provider credibility indicators in the recommendation process. Traditional search interfaces may present providers based primarily on text relevance or predefined filters. The proposed framework introduces provider ratings, successful booking outcomes, interaction frequency, and available user history as additional signals that can influence recommendation ranking.

The framework also generates a ranked list of the Top-20 candidate providers. This output is designed to support subsequent processing by other intelligent components of the complete platform, including temporal and contextual matching, provider skill verification, and review-related analysis. Consequently, the recommendation engine serves as an initial intelligent candidate-ranking component within the wider system.

From a software engineering perspective, the modular architecture is another important product contribution. The recommendation logic is separated into distinct processing stages, allowing individual algorithms to be maintained, tested, or replaced independently. This design supports future improvements such as alternative semantic models, revised credibility indicators, and different hybrid weighting strategies.

The application uses Firebase as its data layer for storing and retrieving the information required by the platform. The recommendation framework is designed to operate alongside the application backend and other intelligent modules, providing a foundation for integration into a complete service provider matching solution.

The practical value of the product lies in its potential to reduce the effort required to identify relevant providers and to introduce additional decision-support information into provider selection. However, the current research does not establish a measured reduction in search time, increased booking completion rates, or improved customer satisfaction. These outcomes require further validation through real-world usage and stakeholder testing.

7.5 Limitations and threats to validity

7.5.1 Category-Based Ground Truth

The primary limitation concerns the definition of recommendation relevance. In the evaluation, a provider was considered relevant when the provider's service category matched the category associated with the user request. This approach provides a consistent and reproducible relevance criterion, but it does not fully represent real-world provider suitability.

A provider belonging to the correct service category may still be unsuitable because of limited experience, poor service quality, unavailability, geographical distance, or other practical constraints. Therefore, the reported Precision@K and NDCG@K results primarily measure category-level recommendation relevance rather than comprehensive provider suitability.

7.5.2 Limited Evaluation of Credibility

The credibility-based collaborative filtering component was designed to represent provider reliability and historical personalization. However, the primary evaluation metrics were based on service-category relevance.

As a result, the evaluation does not directly establish whether the credibility component improves booking success, customer satisfaction, or selection of more trustworthy providers. The low standalone collaborative filtering performance may partly reflect this difference between the component's intended purpose and the evaluation criterion.

Future research should evaluate credibility-related outcomes separately, using indicators such as successful booking rates, verified service completion, and user satisfaction after service delivery.

7.5.3 Cold-Start and Data Sparsity

The collaborative filtering component depends partly on historical user-provider interactions. New users and newly registered service providers may have limited or no interaction history, reducing the amount of information available for personalization.

Although TF-IDF and BERT can generate recommendations using provider descriptions without requiring historical interactions, the credibility and personalization component may remain less informative in cold-start situations. Neutral handling of missing values can reduce extreme scoring behaviour but does not completely solve the cold-start problem.

7.5.4 Dataset and Generalizability Limitations

The experimental evaluation was conducted using the available provider, request, and interaction datasets. A sample of 500 user requests was selected using a fixed random seed to support consistent evaluation.

However, the findings may vary when the framework is applied to different datasets, service categories, languages, geographical regions, or patterns of user behaviour. The results therefore cannot automatically be generalized to all service marketplaces or real-world deployment environments.

Further evaluation using independently collected real-world service requests and provider records would strengthen the external validity of the findings.

7.5.5 Heuristic Weight Selection

The credibility score uses predefined weights for provider rating, booking success rate, and interaction frequency. These weights were selected according to domain-informed reasoning rather than learned through a supervised optimization process.

Similarly, the final hybrid score uses a predefined weighted combination of TF-IDF, BERT, and collaborative filtering. Although weight sensitivity analysis was conducted, the selected configuration may not be optimal for every service category or user group.

Future improvements could investigate data-driven weight optimization, adaptive weighting, or learning-to-rank techniques to determine suitable component contributions under different recommendation conditions.

7.5.6 Potential Bias in Historical Information

Provider ratings and historical interactions may contain biases. For example, highly active providers may accumulate more ratings and interactions, while new providers may have limited opportunities to establish credibility.

Customer ratings may also reflect subjective preferences or inconsistent evaluation behaviour. Consequently, a credibility score derived from historical data should not be treated as an objective or complete measurement of provider quality.

Fairness-aware ranking strategies and periodic monitoring would be useful for identifying whether particular groups of providers are systematically disadvantaged.

7.5.7 Limited Real-World User Validation

The research primarily focused on algorithmic performance and technical implementation. The available evaluation results do not directly measure whether users prefer the generated recommendations or whether the recommended providers successfully complete the requested services.

A formal user study involving service customers and providers would provide additional evidence regarding usability, trust, perceived relevance, and overall satisfaction.

Therefore, the current findings should be interpreted as evidence of experimental recommendation performance and technical feasibility rather than a complete validation of real-world marketplace effectiveness.

7.6 Implications

The findings of this research have several implications for recommendation system development, service marketplace applications, and future research.

From a technical perspective, the results demonstrate the value of combining different recommendation signals within a unified framework. Lexical similarity, semantic similarity, and provider credibility represent different aspects of the recommendation problem. Integrating these signals can provide a more comprehensive ranking strategy than relying on one technique alone.

The results also emphasize the importance of selecting evaluation metrics that reflect the intended purpose of each component. Category-based Precision and NDCG are useful for measuring service relevance, but they do not fully evaluate provider reliability or personalization quality. Therefore, future recommendation research should consider multiple evaluation dimensions, including relevance, credibility, fairness, and real-world service outcomes.

For service marketplace development, the proposed framework provides a foundation for intelligent provider discovery. Instead of requiring users to manually inspect large numbers of provider profiles, a recommendation engine can generate a smaller ranked candidate list based on service-related information and available historical evidence.

The credibility-based collaborative filtering component also highlights the importance of distinguishing service relevance from provider reliability. A provider may offer the correct service but have limited evidence of successful service delivery. Conversely, a highly credible provider may not match the current service requirement. Effective service provider matching should therefore consider both dimensions.

From a product perspective, the modular architecture supports integration with additional intelligent components. Contextual suitability, availability, geographical information, skill verification, and review analysis can complement the recommendation engine to support more informed provider selection.

The research also suggests opportunities for further investigation. Future work could explore adaptive hybrid weighting, improved cold-start handling, learning-to-rank models, fairness-aware recommendations, and evaluation using verified booking outcomes. Additional studies could examine the effect of multilingual semantic matching across different languages and service categories.

Overall, the research demonstrates that a hybrid recommendation strategy can improve category-level provider matching while providing a foundation for incorporating credibility and personalization information. Its practical effectiveness will depend on the quality of the available data, the reliability of the integrated system, and further validation with real users and service outcomes.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

The main objective of this research was to design, implement, and evaluate a hybrid recommendation framework capable of generating relevant and credibility-informed service provider recommendations from natural-language user requests.

To achieve this objective, three recommendation approaches were integrated into a unified ranking process. TF-IDF was used to identify lexical similarity between user requests and provider descriptions, while multilingual BERT was employed to capture semantic relationships that may not be identified through direct keyword matching. In addition, a customized credibility-based collaborative filtering component was developed to incorporate provider ratings, booking success rates, interaction frequency, and available historical user-provider interactions.

The individual recommendation scores were normalized and combined using a weighted fusion strategy. The final hybrid configuration assigned weights of 0.30 to TF-IDF, 0.35 to multilingual BERT, and 0.35 to credibility-based collaborative filtering. This configuration was examined through weight sensitivity analysis to identify a suitable balance between the different recommendation signals.

The experimental evaluation was conducted using Precision@K and NDCG@K at recommendation list sizes of 5, 10, and 20. The proposed hybrid framework achieved the highest overall performance among the evaluated methods. In particular, it achieved a Precision@5 of 0.9490 and an NDCG@5 of 0.9496. At K=20, the framework maintained a Precision@20 of 0.9442 and an NDCG@20 of 0.9458.

These findings indicate that integrating lexical, semantic, and credibility-related information can improve category-level service provider recommendation performance compared with the individual approaches evaluated in this research. The results also demonstrate that TF-IDF and multilingual BERT are effective for identifying service-related relevance, while credibility-based collaborative filtering contributes a different type of information based on provider reliability indicators and historical interactions.

The standalone collaborative filtering component achieved lower category-level relevance performance than the text-based approaches. This finding highlights the importance of combining credibility information with direct service relevance rather than relying on provider credibility alone. It also suggests that credibility-related benefits should be evaluated using additional real-world measures, including booking success and customer satisfaction.

Overall, the research achieved its primary goal of developing and experimentally evaluating a modular hybrid recommendation framework for intelligent service provider matching. The resulting component provides a foundation for generating ranked provider candidates within a broader intelligent service provider matching platform.

However, the evaluation used service-category matching as the primary ground-truth relevance criterion. Therefore, the reported performance values should be interpreted as evidence of category-level recommendation relevance rather than proof of real-world provider suitability or customer satisfaction.

8.2 Achievement of objectives

The research objectives established at the beginning of the study were addressed through the design, implementation, and experimental evaluation of the proposed framework.

The first objective was to develop a recommendation approach capable of identifying service providers relevant to natural-language user requests. This objective was addressed by implementing TF-IDF and multilingual BERT matching techniques. TF-IDF supported direct keyword-level relevance, while BERT provided semantic matching capabilities for differently worded service descriptions.

The second objective was to incorporate provider credibility and historical interaction information into the recommendation process. This objective was addressed through the development of a customized credibility-based collaborative filtering component. The component combined provider ratings, booking success rates, interaction frequency, and available previous user-provider interactions to produce an additional recommendation signal.

The third objective was to combine the individual recommendation techniques into a unified hybrid framework. This was achieved through score normalization and weighted fusion. The framework combined lexical, semantic, and credibility-related scores using the selected weight configuration and generated a ranked list of the Top-20 candidate service providers.

The fourth objective was to evaluate the effectiveness of the hybrid recommendation framework against the individual recommendation approaches. This objective was achieved through comparative evaluation using Precision@K and NDCG@K at K values of 5, 10, and 20. The results demonstrated that the hybrid approach achieved the highest category-level recommendation performance among the evaluated methods.

The fifth objective was to support integration of the recommendation framework into the complete service provider matching platform. The component was designed to produce ranked provider candidates that could be processed by other intelligent modules, including contextual matching, provider verification, and review analysis. Firebase was used as the application data layer supporting the broader system.

Table 8.1: Achievement of Research Objectives

Research Objective	**Achievement**	**Evidence**
Develop natural-language provider matching	Achieved	TF-IDF and multilingual BERT components
Incorporate provider credibility	Implemented	Customized credibility-based CF
Develop a hybrid recommendation framework	Achieved	Normalization, weighted fusion, Top-20 ranking
Evaluate recommendation performance	Achieved	Precision@K and NDCG@K results
Support integration with the complete platform	Designed and implemented at	Modular recommendation output and application

	component level	architecture

The findings demonstrate that the primary algorithmic objectives were achieved. However, the integration objective should be interpreted according to the level of system integration completed and verified during implementation. Full real-world validation of the complete platform remains a potential area for further work.

8.3 Individual research and product contributions

The individual contribution of this research focuses on the design, development, implementation, and evaluation of the Hybrid Recommendation Framework, which operates as one intelligent component of the complete service provider matching platform.

The primary research contribution is the integration of three complementary recommendation techniques into a unified framework. Rather than depending exclusively on keyword matching or historical interactions, the proposed approach combines TF-IDF, multilingual BERT, and credibility-based collaborative filtering to represent different aspects of provider matching.

A particularly important individual contribution is the design of the customized credibility-based collaborative filtering component. Traditional collaborative filtering approaches primarily rely on historical interaction patterns. In contrast, the proposed component explicitly incorporates provider credibility indicators, including customer ratings, successful booking outcomes, and interaction frequency.

The credibility indicators were combined using domain-informed weights, with 0.50 assigned to ratings, 0.30 to booking success rate, and 0.20 to interaction frequency. A transformation was applied to interaction frequency to reduce the influence of exceptionally active providers. Previous user-provider interactions were also considered to introduce a controlled personalization adjustment when historical information was available.

Another research contribution is the implementation of score normalization and weighted fusion to combine the outputs of the three recommendation components.

The selected hybrid configuration was evaluated through weight sensitivity analysis, providing evidence to support the balance among lexical, semantic, and credibility-related information.

The individual experimental contribution includes evaluating TF-IDF, BERT, credibility-based collaborative filtering, and the hybrid approach under a consistent category-based relevance protocol. The analysis included recommendation performance comparison, score distribution examination, component correlation analysis, and weight sensitivity evaluation.

From a product development perspective, the contribution includes implementing a modular recommendation pipeline that processes natural-language service requests and generates ranked Top-20 provider candidates. The component was designed to communicate with the surrounding application architecture and support subsequent processing by other intelligent modules.

The work also contributed implementation artefacts, including Python-based recommendation code, Google Colab experiments, preprocessing procedures, evaluation scripts, performance results, and supporting visualizations. These artefacts provide a foundation for future maintenance, experimentation, and further development of the recommendation component.

The contribution should be understood as an application-specific extension and integration of established recommendation techniques rather than the invention of a completely new foundational machine learning algorithm. Its novelty lies in the customized credibility-aware scoring design, the combination of complementary recommendation signals, and their application to intelligent service provider matching.

8.4 Future directions

8.4.1 Adaptive Weight Optimization

The current framework uses predefined weights for combining TF-IDF, multilingual BERT, and credibility-based collaborative filtering. Although weight sensitivity analysis supported the selected configuration, fixed weights may not be optimal for every service category or user requirement.

Future research could investigate adaptive weight optimization techniques that automatically adjust component contributions based on the characteristics of a request, the available interaction history, or observed recommendation outcomes. Learning-to-rank approaches could also be explored to learn effective ranking combinations directly from labelled interaction data.

8.4.2 Improved Cold-Start Handling

Cold-start situations remain a challenge when new users or providers have limited historical interaction information. Although TF-IDF and BERT can generate content-based recommendations without relying on previous interactions, the credibility-based collaborative filtering component may have insufficient information for personalization.

Future improvements could incorporate additional provider profile attributes, verified qualifications, service experience, and other relevant information to support the ranking of newly registered providers. New-user preferences could also be collected through optional onboarding questions or initial service requirements.

8.4.3 Real-World User and Provider Validation

The current evaluation primarily measures service-category relevance using Precision@K and NDCG@K. Future work should extend the validation to real users and service providers to examine the practical usefulness of the recommendations.

User studies could evaluate perceived relevance, ease of provider selection, trust in recommendation results, and overall satisfaction. Real-world application data could also be used to examine booking completion rates, repeated provider selection, and successful service delivery.

Such validation would provide stronger evidence of whether the credibility-based collaborative filtering component improves practical provider selection beyond category-level matching.

8.4.4 Enhanced Credibility and Trust Modelling

The existing credibility component uses provider ratings, booking success rate, and interaction frequency. Future versions could incorporate additional reliability

indicators such as verified professional qualifications, service completion consistency, complaint history, and verified customer feedback.

The framework could also investigate methods for detecting unreliable ratings or unusual interaction patterns that may distort provider credibility scores. Integration with review credibility analysis could further improve the reliability of the information used for ranking.

8.4.5 Multilingual and Context-Aware Recommendation

The current framework uses a pretrained multilingual sentence-transformer model for semantic matching. Future research could evaluate its performance more extensively across Sinhala, Tamil, English, and mixed-language service requests.

Additional improvements could include adapting the semantic model using domain-specific service descriptions and real-world user queries. This may help the system better understand informal language, spelling variations, and terminology commonly used in local service marketplaces.

The recommendation framework could also be enhanced through closer integration with contextual matching information, including provider availability, service location, time requirements, and other practical constraints. These factors could help distinguish between providers who are textually relevant and those who are practically suitable for a particular request.

8.4.6 Scalability and Deployment Optimization

As the number of service providers and user requests increases, the computational cost of generating recommendations may also increase. Future development could investigate more efficient candidate retrieval, precomputed provider embeddings, caching, and optimized ranking procedures.

The system could also benefit from operational monitoring to measure recommendation response time, application reliability, and data-processing performance under realistic workloads. These improvements would support deployment in a larger service marketplace environment.

8.4.7 Fairness and Explainable Recommendations

Future research could investigate fairness-aware recommendation techniques to reduce the risk of systematically disadvantaging new or less active service providers. Additional monitoring could help identify whether ranking outcomes are disproportionately influenced by historical popularity or incomplete provider information.

Explainable recommendation features could also be introduced to help users understand why particular providers were recommended. For example, the application could display concise explanations based on service relevance, semantic matching, and verified credibility indicators. Such explanations could improve transparency and support more informed provider selection.

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10 GLOSSARY

Hybrid Recommendation Framework - A recommendation approach that combines multiple techniques to produce a unified ranking of service providers.

Lexical Similarity - The degree of similarity between texts based on shared words or terms.

Semantic Similarity - The degree to which two texts express similar meanings, even when different words are used.

Credibility Score - A numerical value representing provider-related reliability indicators, including ratings, booking success, and interaction frequency.

Collaborative Filtering - A recommendation technique that uses historical user-item or user-provider interactions to support personalized recommendations.

Cold-Start Problem - A recommendation challenge that occurs when a new user or provider has insufficient historical information.

Data Sparsity - A condition where only a small proportion of possible user-provider interactions have been recorded.

Sentence Embedding - A numerical vector representation of a sentence that captures aspects of its meaning.

Cosine Similarity - A measure used to compare the similarity of two numerical vectors based on their direction.

Score Normalization - The process of transforming scores into a common numerical range so they can be compared or combined.

Weighted Fusion - A method of combining several scores by assigning a specified contribution to each score.

Ground Truth - Reference information used to determine whether a recommendation is relevant during evaluation.

Category-Level Relevance - A relevance criterion where a recommended provider is considered relevant when its service category matches the category of the user request.

Ranking Quality - The effectiveness of ordering relevant providers toward the highest positions in a recommendation list.

Top-K Recommendation - The highest-ranked K items or providers selected from a recommendation list

Weight Sensitivity Analysis - An analysis of how changes in component weights affect recommendation performance.

Provider Credibility - The perceived reliability of a service provider based on available service-related evidence.

Interaction Frequency - The number of recorded interactions associated with a provider.

11 APPENDICES

11.1 Appendix A Research instruments or extended results

11.1.1 A.1 Dataset Structure and Attributes

The experimental implementation used three primary data groups: provider information, user requests, and historical user-provider interactions. These datasets supported lexical matching, semantic matching, credibility calculation, and recommendation evaluation.

Table 11.1:Main Dataset Attributes

Dataset	Main Attributes	Purpose
Provider Dataset	Provider ID, service category, service description, location, rating, experience	Supports content-based and semantic matching
User Request Dataset	Request text, service category	Provides recommendation queries and evaluation relevance labels
User Interaction Dataset	User ID, provider ID, rating, booking outcome, interaction frequency	Supports credibility scoring and historical personalization

11.1.2 A.2 Extended Experimental Results

The recommendation methods were evaluated using Precision@K and NDCG@K at K values of 5, 10, and 20. The evaluation used 500 randomly selected user requests and category-based relevance labels.

Table 11.2:Detailed Recommendation Performance

Method	P@5	P@10	P@20	NDCG@5	NDCG@10	NDCG@20
TF-IDF	0.9458	0.9378	0.9231	0.9471	0.9412	0.9297
BERT	0.9467	0.9442	0.9387	0.9469	0.9451	0.9410
CF	0.0633	0.0651	0.0672	0.0613	0.0633	0.0654
Hybrid	0.9490	0.9483	0.9442	0.9496	0.9490	0.9458

The results demonstrate that the hybrid approach achieved the highest category-level recommendation performance among the evaluated methods. However, these results do not directly establish real-world customer satisfaction or provider reliability.

11.1.3 A.3 Research Artefacts

The research produced several technical artefacts supporting implementation, experimentation, and evaluation. These include the Python-based recommendation implementation, Google Colab notebook, dataset preprocessing procedures, scoring and ranking logic, evaluation scripts, experimental result tables, and supporting visualizations.

The Google Colab notebook provides a record of the main experimental workflow, including dataset preparation, recommendation score generation, hybrid fusion, and evaluation. The artefacts may be used to reproduce the experiments under comparable dataset and configuration conditions.

Where source code or datasets cannot be shared publicly because of privacy, access, or institutional restrictions, their availability should be described accurately rather than assumed.

11.2 Appendix B Academic integrity and AI use disclosure

11.2.1 B.1 Declaration of Academic Integrity

I declare that the research, implementation, experimental evaluation, and findings presented in this thesis represent my own academic work and individual contribution to the Hybrid Recommendation Framework for Intelligent Service Provider Matching, except where external sources or contributions have been appropriately acknowledged.

Established algorithms, libraries, research publications, and supporting resources used during the research have been identified and referenced where applicable. The reported experimental results are based on the evaluation conducted for the developed recommendation framework.

I acknowledge that artificial intelligence tools were used as supporting resources during the preparation of this thesis, as described in the following sections. Responsibility for the accuracy, originality, interpretation, and final presentation of the submitted work remains with me.

11.2.2 B.2 AI Tools Used

Table 11.3:AI Tool Usage Disclosure

AI Tool	Purpose	Affected Work
ChatGPT (OpenAI)	Assistance with academic writing, language refinement, and document organization	Selected thesis sections and supporting documentation

11.2.3 B.6 Limitations of AI Assistance

Generative AI tools may produce inaccurate technical explanations, unsupported claims, or content that does not fully reflect the implemented system. Therefore, AI assistance was used only as a supporting tool for documentation and communication.

The research conclusions are based on the experimental evaluation and implementation evidence rather than on AI-generated assertions. The use of AI

assistance does not replace the requirement for appropriate citations, independent verification, academic integrity, or compliance with institutional research policies.